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Optimal Inventory Management using Model Predictive Control in Wastewater Treatment Plants

Master's thesis in Industrial Chemistry and Biotechnology

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June 2026

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ABSTRACT

Wastewater treatment plants are complex process systems that require effective control systems to ensure reliable operation and that the effluent quality meets required standards. Model predictive control is suitable for such systems due to its ability to handle multivariable dynamics and operational constraints within a unified optimization framework. Recent research has demonstrated an MPC scheme for optimal inventory control for complex process topologies. However, these approaches have not accounted for reaction dynamics.

This thesis investigates the application of nonlinear model predictive control for optimal inventory management in a simplified activated sludge process. A dynamic process model was developed, incorporating volume dynamics, reaction dynamics, component balances, and a recycle stream. The model was integrated into an NMPC framework using orthogonal collocation on finite elements. The controller objective was to balance throughput maximization and inventory management, while satisfying process constraints.

A series of case studies with increasing process complexity was conducted to evaluate controller performance under flow disturbances and variations in influent substrate concentration. The results demonstrate that the proposed NMPC formulation successfully manages inventories. The controller effectively exploits available buffer capacities to minimize the cost function and satisfy process constraints.

The main contribution of this work is the extension of an existing optimal inventory control MPC framework to a wastewater treatment process containing reaction dynamics, component balances, and a recycle stream. Although several simplifying assumptions were made regarding the process and models, the results indicate that the proposed approach is a step toward applying the NMPC scheme in real applications. Future work should focus on more detailed process models, economic optimization, state estimation, and evaluation under realistic operating conditions.

ABSTRAKT

Avløpsrenseanlegg er komplekse prosesssystemer som krever effektive kontrollsystemer for å sikre stabil drift og at kvaliteten på det rensede avløpsvannet oppfyller gjeldende krav. Modellprediktiv regulering (MPC) er særlig attraktiv for slike systemer på grunn av evnen til å håndtere multivariable dynamikker og operasjonelle beskrankninger innenfor ett samlet optimaliseringsrammeverk. Tidligere forskning har demonstrert et MPC-rammeverk for optimal inventarstyring (inventory control) i komplekse prosesstopologier. Disse tilnærmingene har imidlertid ikke tatt hensyn til reaksjonsdynamikk.

Denne oppgaven undersøker anvendelsen av ikke-lineær modellprediktiv regulering (NMPC) for optimal inventarstyring i en forenklet aktiv slam prosess. Det ble utviklet en dynamisk prosessmodell som inkluderer volumdynamikk, reaksjonsdynamikk, komponentbalanser og en resirkuleringsstrøm. Modellen ble integrert i et NMPC-rammeverk ved bruk av ortogonal kollokasjon på endelige elementer som diskretiseringsmetode. Regulatorens mål var å balansere maksimal gjennomstrømning og inventarstyring samtidig som prosessbeskrankninger ble overholdt.

En serie casestudier med økende prosesskompleksitet ble gjennomført for å evaluere regulatorens ytelse under strømningsforstyrrelser og variasjoner i innløpskonsentrasjonen av substrat. Resultatene viser at den foreslåtte NMPC-formuleringen håndterer inventarene på en tilfredsstillende måte. Regulatoren utnytter tilgjengelig bufferkapasitet effektivt for å minimere kostfunksjonen og samtidig respektere prosessbeskrankningene.

Hovedbidraget i denne oppgaven er utvidelsen av et eksisterende MPC-rammeverk for optimal inventarstyring til en avløpsrenseprosess som inkluderer reaksjonsdynamikk, komponentbalanser og en resirkuleringsstrøm. Selv om flere forenklinger ble gjort i prosessmodellen, indikerer resultatene at det foreslåtte rammeverket representerer et steg mot anvendelse i reelle industrielle prosesser. Videre arbeid bør fokusere på mer detaljerte prosessmodeller, økonomisk optimalisering, tilstandsestimering og evaluering under mer realistiske driftsforhold.

PREFACE

This master's thesis was written during the spring semester of 2026 as part of the Master of Science program in Industrial Chemistry and Biotechnology at Norwegian University of Science and Technology (NTNU). This work was conducted at the Department of Chemical Engineering within the Process Systems Engineering group. The thesis investigates optimal inventory control with model predictive control, in a wastewater treatment process.

I would like to express my sincere gratitude to my supervisor, Johannes Jäschke, for guidance and constructive feedback, throughout this project. I would also like to express my gratitude to my co-supervisor, Archana Kumaraswamy, for her continuous support throughout this project. Her willingness to always make time for questions and discussions, and her constant encouragement have been invaluable during this work. Thank you!

CONTENTS

Abstract	i
Abstrakt	i
Preface	ii
Contents	iv
List of Figures	iv
List of Tables	vi
Abbreviations	viii
1 Introduction	1
1.1 Background	1
1.2 Motivation	1
1.3 Previous Work and Scope	2
1.4 Thesis Structure	3
2 Theoretical Background	5
2.1 Wastewater Treatment plant	5
2.1.1 Activated Sludge Process	5
2.1.2 Activated Sludge Model	6
2.2 Control Formulation	8
2.2.1 Inventory Control	8
2.2.2 Model Predictive Control	9
2.2.3 Discretization Method	11
3 Methods	13
3.1 Dynamic Model	13
3.1.1 Volume Dynamics	13
3.1.2 Biochemical Reaction Dynamics	14

3.1.3	Component Balances	15
3.1.4	Settling Tank Dynamics	16
3.2	Control Formulation	17
3.2.1	Development of the MPC Formulation	17
3.2.2	The Volumetric Mass Transfer Coefficient, K_La	19
3.2.3	Return Activated Sludge	19
3.2.4	MPC Formulation	20
3.2.5	Closed-Loop Simulation Disturbances	21
4	Case Studies	23
4.1	Case 1: Single Bioreactor Tank	23
4.2	Case 2: Bioreactor Incorporated into a Sequential Tank System	26
4.3	Case 3: Sequential Tank System with Substrate Perturbation	29
4.4	Case 4: Activated Sludge Process	32
4.4.1	Flow disturbance	33
4.4.2	Influent substrate concentration perturbation	38
5	Discussion and Future Work	43
6	Conclusions	47
	References	49
	Appendices:	i
	A - Github repository	ii
	B - Simulation of bioreactor with RK4	iii

LIST OF FIGURES

2.1.1 ASP configuration	6
2.2.1 MPC Strategy Image courtesy of Martin Behrendt.[33].	10
2.2.2 Basic structure of MPC. Figure inspired by Camacho et al. [32]	10
3.1.1 Settling tank model.	17
4.1.1 Block diagram of single bioreactor system	24
4.1.2 Flow rates in Single Bioreactor system.	24
4.1.3 Volume in bioreactor	25
4.1.4 State variables in bioreactor	25
4.2.1 Block diagram of bioreactor in a sequential arrangement.	27
4.2.2 Flow rates.	27
4.2.3 Volumes.	28
4.2.4 State variables, X_B , S_S , S_O	29
4.3.1 Block diagram of bioreactor in a sequential arrangement, with substrate perturbation.	30
4.3.2 Flow rates, influent substrate concentration profile, and K_La profile.	31
4.3.3 Volumes.	31
4.3.4 State variables, X_B , S_S , S_O	32
4.4.1 Block diagram of activated sludge process.	33
4.4.2 Flow rates.	34
4.4.3 Constant inlet substrate concentration.	34
4.4.4 Tank volumes and bioreactor states.	34
4.4.5 K_La and recycle ratio.	35
4.4.6 Substrate and biomass concentrations in the tanks.	35
4.4.7 Flow rates.	38
4.4.8 Inlet substrate concentration.	38
4.4.9 Tank volumes and bioreactor states.	39
4.4.10 K_La and recycle ratio.	39
4.4.11 Substrate and biomass concentrations in the tanks.	40

.0.1	Components in bioreactor.	iii
.0.2	Flow rates.	iii

LIST OF TABLES

2.1.1 Stoichiometric matrix for biomass growth and decay.	7
3.2.1 Model parameters and typical domestic wastewater characteristics (Denmark, 20°C)	22
3.2.2 Inlet values	22

ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **ASM1** Activated Sludge Model No. 1
- **ASP** Activated Sludge Process
- **DAE** Differential Algebraic Equation
- **MPC** Model Predictive Control
- **NLP** Nonlinear Programming
- **NMPC** Nonlinear Model Predictive Control
- **OCFE** Orthogonal Collocation on Finite Elements
- **RAS** Return Activated Sludge
- **TPM** Throughput Manipulator
- **WAS** Waste Activated Sludge
- **WWTP** Wastewater Treatment Plant

INTRODUCTION

1.1 Background

Wastewater treatment plants (WWTPs) are critical infrastructure for protecting aquatic ecosystems and maintaining water quality by reducing the discharge of nutrients and pathogens into receiving waters [1, 2].

Currently, the marine ecosystem of the Oslofjord is under significant pressure [3]. One explanation for this situation is excessive nutrient loading. The release of excess nutrients into oceans may lead to harmful algal blooms, which further deplete dissolved oxygen concentrations. This process is known as eutrophication. Under hypoxic or anoxic conditions, normal ecosystem functioning is disrupted [4]. Municipal wastewater discharge is one of the main drivers towards deterioration [2, 5].

To remediate the marine ecosystems, and prevent further deterioration, the Norwegian government has introduced new restrictions on WWTPs [6]. The new regulations require agglomerations exceeding 1000 PE to implement at least secondary treatment. Primary treatment includes mechanical removal of objects, solids and particles, whereas secondary treatment comprises chemical or biological processes that reduce organic matter content in municipal wastewater [6]. The upgrade of existing WWTPs and installation of new facilities require effective and reliable control systems.

1.2 Motivation

WWTPs typically consist of interconnected process units including primary settlers, biological reactors, aeration tanks, secondary clarifiers and recycle streams [7]. As a result, WWTPs form complex process topologies, and large dynamic systems with multiple interacting inventories and operational constraints. To ensure reliable operation, a control system with effective inventory management is required. Managing material balances

during operation is important for operational safety, product quality and profitability [8]. In addition, the control system should ensure that the effluent quality meets required standards. This implies that operational conditions for biological pollutant removal must be controlled such that effluent concentrations of pollutants are within limits. This is a challenge because tank inventories in biological treatment systems are coupled with reaction dynamics. Hence, changes in flow and inventory levels influence both process economics and treatment performance. These interactions introduce multivariable dynamics and operational constraints, making model predictive control (MPC) attractive [9]. Model predictive control (MPC) is particularly well suited for this task, as it enables simultaneous handling of multivariable dynamics and operational constraints [10].

1.3 Previous Work and Scope

Recent work by Turan et al. [11] and Kumaraswamy et al. [12] demonstrates an MPC framework to achieve optimal inventory control in complex process topologies. Inventories are represented by tank liquid levels and controlled through systematic weighting of inventories and system outflows in the MPC cost function. The MPC design assumes that throughput maximization is equivalent to maximizing profit. The objective is to isolate bottlenecks and maximize throughput while maintaining feasible operation [11, 12].

In the project thesis [13], this framework was extended by modelling liquid tanks with a mixing process. The project thesis investigated the implementation of soft constraints on component concentrations within the tanks. For simplification, only one component was explored with 3 tanks in parallel and sequential arrangement. The non-linear model predictive controller in the project thesis was implemented using the Euler discretization method on the component balance.

In this master thesis, we seek to further extend the application of the inventory control framework to a realistic process. Here, we consider a simplified secondary treatment process in a WWTP. Firstly, a dynamic model is developed for a simplified activated sludge process (ASP), including a mixing tank with buffer capacity, an aerated bioreactor, a settling tank, and a return activated sludge stream. The simplified model demonstrates how biomass and substrate concentrations, tank volumes, and dissolved oxygen content evolves with time. Secondly, the dynamic model is used to build a non-linear model predictive controller using orthogonal collocation as a discretization method. Orthogonal collocation is selected because it provides an efficient transcription of the nonlinear dynamic model, and is well suited for systems with strongly coupled and potentially stiff dynamics. Lastly, we have incorporated the optimal inventory control formulation from Kumaraswamy et al. [12] within the NMPC framework for the ASP. Here, inventories include both tank volumes and concentration of components. The control objectives are as follows. Firstly, we seek to maintain substrate concentration in the bioreactor effluent

below a specified limit. Secondly, we assume that the outflow of treated wastewater should be maximized in order to utilize the available treatment capacity. Third, the liquid inventories of the process units must be maintained within their operational limits to avoid overflow and complete depletion. Finally, the return activated sludge flow should be regulated to maintain a sufficient biomass concentration in the bioreactor, thereby ensuring effective biological treatment. The NMPC therefore balances throughput maximization, inventory management, and process constraint satisfaction within a unified optimization framework.

Thereby, this master thesis implements and evaluates the optimal inventory control scheme in a more complex and realistic process that includes reactions, component balances, and a recycle. The primary contribution of this work is the adaptation of the existing optimal inventory control MPC framework to an activated sludge process, enabling it to satisfy process constraints while preserving optimal inventory management.

1.4 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 presents the theoretical background relevant to this work, including the activated sludge process, inventory control, and model predictive control. Chapter 3 describes the dynamic process model and the MPC formulation developed in this study. Chapter 4 presents the case studies and discusses the corresponding simulation results. Chapter 5 provides a discussion of the findings, limitations, and opportunities for future work. Finally, Chapter 6 summarizes the main conclusions of the thesis.

THEORETICAL BACKGROUND

In this chapter, the theoretical background relevant to this master's thesis is presented. The chapter is divided into two main parts: wastewater treatment plant theory and control formulation. Sections 2.1.1 and 2.1.2 outline the operating principles of a secondary wastewater treatment process and present a mathematical model describing this process. Sections 2.2.1 and 2.2.2 discuss the theoretical background of inventory control and provide an overview of the principles of model predictive control. Lastly, we review the principles of orthogonal collocation.

2.1 Wastewater Treatment plant

2.1.1 Activated Sludge Process

Municipal wastewater may contain various pollutants. A WWTP is an industrial facility designed to remove pollutants in the influent wastewater. The treatment strategies may vary depending on wastewater properties, but WWTPs typically consist of a combination of mechanical, physical, chemical and biological processes [7]. The activated sludge process (ASP) is the predominant biological secondary treatment for municipal wastewater [14]. The general process flow is that influent wastewater enters an aerated bioreactor, followed by a settling tank. From the settling tank the clarified water is discharged or sent for further treatment. Part of the sludge at the bottom of the settling tank is recycled back to the bioreactor to maintain a high biomass concentration [7]. This is called return activated sludge (RAS). The remaining sludge is further treated and disposed, and is called waste activated sludge (WAS). This process configuration is depicted in Figure 2.1.1

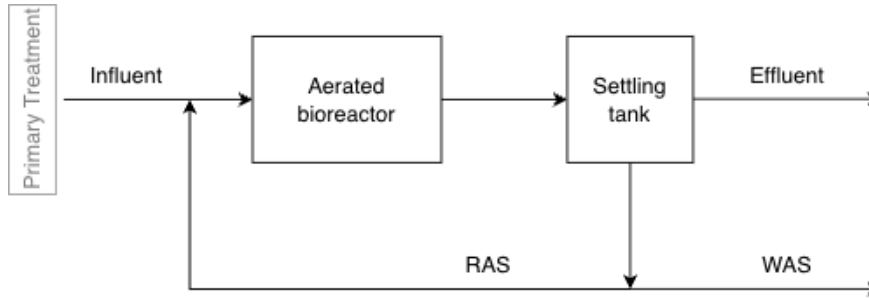


Figure 2.1.1: ASP configuration

In the bioreactor there exist microorganisms, such as bacteria. Heterotrophic bacteria feed on organic matter to perform metabolism and microbial growth [15, 7]. This mechanism, aerobic biodegradation, is exploited in the technology to achieve organic matter removal. To ensure optimal growth conditions, the tank is aerated. In the settling tank, treated water and the activated sludge, which is flocculated biomass, are separated by gravitational forces [7].

2.1.2 Activated Sludge Model

In 1987, a task group formed by IAWPRC published the Activated Sludge Model no.1 (ASM1) [16]. The ASM1 has been a fundamental part of the modeling of biological wastewater treatment [17, 18, 19]. The activated sludge process is highly complicated to model, as it is a large non-linear system, subject to significant perturbations from the influent [20]. The ASM1 provides the simplest model having the capability of realistic predictions of the performance of single sludge systems carrying out carbon oxidation, nitrification and di-nitrification [16]. The model consists of 13 components and 8 processes. For further details on the model, see Henze et al. [16].

We can further simplify the full ASM1 model into a model consisting of only 3 key components: biomass, substrate, and dissolved oxygen. This simplification is based on the assumption that heterotrophic bacteria is the only active organism. In addition, we assume that the heterotrophic bacteria only utilize soluble substrate as carbon source, and that this occur in an aerobic environment. In this simplified model, autotrophic biomass and its related processes are disregarded, as well as nitrification and di-nitrification processes. Hence, the governing processes are bacterial growth and decay. The substrate consumption and oxygen utilization is determined through stoichiometric relations to the governing processes. The conversion rates for each component in these processes are collected in Table 2.1.1 [16].

Table 2.1.1: Stoichiometric matrix for biomass growth and decay.

Process, j		Component, i			Process rate, ρ_j
		X_B	S_S	S_O	
1	Growth	1	$-\frac{1}{Y}$	$-\frac{1-Y}{Y}$	$\frac{\hat{\mu}S_S}{K_S + S_S}X_B$
2	Decay	-1	0	-1	bX_B

Observed conversion rates $r_i = \sum_j r_{ij} = \sum_j \nu_{ij}\rho_j$

The stoichiometric matrix connects processes, j (rows), to components, i (columns), via coefficients ν_{ij} . The model uses Monod kinetics for biomass growth and a first-order Herbert expression for biomass decay. The specific growth rate using Monod kinetics is

$$\mu = \frac{\hat{\mu}S_S}{K_S + S_S} \quad (2.1)$$

where $\hat{\mu}$ is the maximum specific growth rate of the heterotrophic biomass. K_S is the half-velocity constant for the readily biodegradable substrate, defined as the substrate concentration at which the growth rate equals half of maximum rate [21]. The parameter b in ρ_2 is the first-order decay coefficient for heterotrophic biomass.

The resulting observed conversion rates, r_i for insoluble biomass (X_B), soluble substrate (S_S) and dissolved oxygen (S_O) become

$$\begin{aligned} r_{X_B} &= \left(\frac{\hat{\mu}S_S}{K_S + S_S} - b \right) X_B \\ r_{S_S} &= -\frac{1}{Y} \cdot \frac{\hat{\mu}S_S}{K_S + S_S} X_B \\ r_{S_O} &= -\frac{1-Y}{Y} \cdot \frac{\hat{\mu}S_S}{K_S + S_S} X_B - bX_B \end{aligned} \quad (2.2)$$

Y is the true growth yield, defined as the mass of heterotrophic biomass produced per unit mass of substrate consumed. It quantifies the efficiency of substrate utilization for biomass synthesis, and takes values in the range $0 < Y < 1$.

In this simple model, the growth process is proportional to biomass concentration in a first order manner and to substrate in a mixed order manner, while decay is proportional in a first order manner with respect to biomass [16]. In the reaction rate for biomass in Equation 2.2, the oxygen dependence is not included. In other words, if the dissolved oxygen concentration goes towards zero, the biomass may still grow if substrate is present.

The stoichiometric matrix only describes the internal conversion terms, and has to be combined with advective terms to describe the component dynamics in a reactor. The general mass balance is given in Equation 2.3 [22]. The conversion rates should be implemented in the *Generated – Lost* term, with *In* and *Out* terms consistent with the system.

$$\underbrace{\text{Change in inventory}}_{\substack{\text{accumulated} \\ \text{within system}}} = \underbrace{\text{In} - \text{Out}}_{\substack{\text{through} \\ \text{system boundary}}} + \underbrace{\text{Generated} - \text{Lost}}_{\substack{\text{internally} \\ \text{in system}}}. \quad (2.3)$$

2.2 Control Formulation

2.2.1 Inventory Control

Inventory control is an important aspect in control strategies, as inventories are present in all process plants, such as tank levels, temperature, etc [23]. Optimal control is essential to achieve reliable and safe operation, while optimizing economics. Inventory control is equivalent to material balance control, as the control system seeks to adjust flows so that steady-state is maintained over the system, which prevents undesired operation, such as accumulation or total depletion of inventories [23].

There are in general two types of inventory control strategies: tight level control and averaging level control [9]. In the case where a buffer tank is implemented in a process system to dampen disturbances for subsequent unit(s), it is desirable to use averaging level control. Two fundamental disturbances are quality (temperature, concentration, etc.) and flow rate [24]. In the case of frequent concentration variations in the influent, the buffer tank is used to average the concentration before the fluid is sent to the subsequent process units. In the case of frequent flow disturbances, the buffer tank is used to dampen the flow rate fluctuations to downstream vessels. In contrast, tight level control aims to exploit the buffer tank capacity in order to allow for independent unit operations under, e.g., downtime. In such case, the level setpoint is set close to maximum or minimum level.

Several studies have worked with inventory control based on tight level control [25, 26]. These studies consider serially connected tanks and exploits the storage capacities to maximize throughput of the system by isolating bottlenecks. Consider such a serially connected tank system, where the throughput manipulator (TPM, the degree of freedom used to set the steady-state flow in the system [27]) is located at the outflow. If an upstream disturbance introduce a bottleneck in the inflow, maintaining all tank levels at fixed setpoints would force the downstream flow rates to decrease, thereby reducing the overall throughput. However, by exploiting the buffer capacities of the tanks, the disturbance propagation can be delayed. Rather than reducing downstream flow rates immediately, the flows downstream of the bottleneck continue operating at their maximum feasible rates. As a result, the inventory in the first tank is depleted until its level

reaches the minimum limit. Subsequently, the available inventory in each downstream tank is utilized in turn, allowing the maximum outflow to be maintained until all buffer capacities have been depleted. The studies propose a decentralized control strategy, specifically bidirectional control, in order to address this control philosophy. The strategy is based on using simple controllers on both inflow and outflow of each unit with setpoints of maximum and minimum limits. The control loops rearrange automatically using selectors. For further details on the bidirectional control strategy, the reader is referred to Skogestad [9].

Turan et al. [11] has proposed an MPC formulation equivalent to bidirectional control. The MPC scheme isolates bottlenecks and maximizes throughput for serially connected tanks, by exploiting the buffer capacities of the tanks. Kumaraswamy et al. [12] has further adapted the MPC scheme to more complex process topologies, including flow splits, mergers and recycles. The work show how selected inventories or system outflows can be prioritized, and present a systematic procedure to determine inventory weights.

2.2.2 Model Predictive Control

MPC is a control strategy that has become widely used in large-scale industrial processes, particularly in systems with multiple interacting inputs and outputs [28, 29]. Following its industrial introduction in the 1970s [30], MPC was regarded as a promising alternative to conventional control approaches because it can explicitly incorporate process dynamics and operational constraints within an optimization framework [9]. Despite these advantages, the widespread adoption of MPC has been hindered by two major challenges [31]. Firstly, effective implementation relies on an accurate process model. Secondly, solving the optimization problem can be a computational burden. Consequently, MPC has not achieved the industrial establishment that some anticipated [9]. Nevertheless, its ability to systematically handle dynamic interactions and constraints makes MPC a powerful tool for inventory management in complex process systems.

The operating principle of MPC is illustrated in Figure 2.2.1. At each time step k , the future behavior of the system is predicted over a finite prediction horizon. The predicted output depends on measured outputs, past control inputs and on the future control signals, which are calculated. The predicted output trajectory is compared to a reference trajectory (e.g. a setpoint), and an optimization problem is solved to determine future control inputs. Only the first control input is implemented. At the next time step, new measurements are obtained and the optimization is repeated, resulting in a receding-horizon strategy [32].

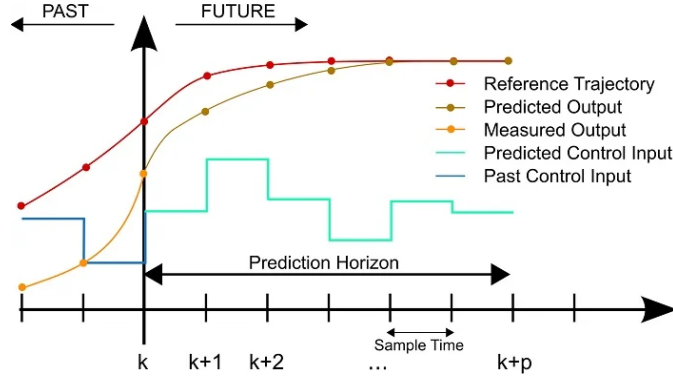


Figure 2.2.1: MPC Strategy Image courtesy of Martin Behrendt.[33].

Figure 2.2.2 shows the basic structure of an MPC controller. Previous measurements together with future control inputs are given to a process model, which calculates predicted outputs. These predictions are compared with the reference trajectory to calculate the expected tracking error. The optimizer then determines the control sequence that minimizes the cost function while satisfying operational constraints. The optimized input sequence is subsequently used to update the controller action.

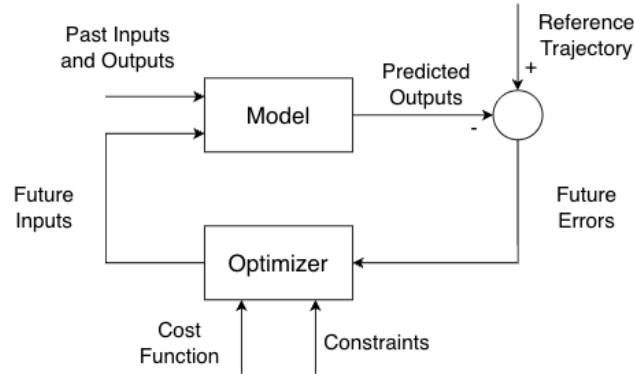


Figure 2.2.2: Basic structure of MPC. Figure inspired by Camacho et al. [32]

A general nonlinear programming (NLP) formulation is

$$\begin{aligned}
 \min_{x,u} \quad & J(x,u) \\
 \text{s.t.:} \quad & f(x,u) = 0 \\
 & h(x,u) = 0 \\
 & g(x,u) \leq 0
 \end{aligned} \tag{2.4}$$

Where J is the cost function, f represents the dynamic model of the system, h represents equality constraints and g represents the inequality constraints. x denotes the state variables, and u denotes input variables (manipulated variables).

A generalized cost function is

$$J = \sum_k^{n_p} (y_k - y_k^{setpoint})^2 \cdot Q + \sum R \quad (2.5)$$

Where the first quadratic term represent state tracking, and R represents regularization terms. In the MPC the NLP is repeatedly solved for each time step.

2.2.3 Discretization Method

The dynamic model is essential in MPC, as the controller computes the optimal control actions based on predicted system behavior [34]. An optimization problem is typically infinite-dimensional, due to such as state trajectories that are continuous functions of time. To solve the MPC problem numerically, the infinite-dimensional system dynamics must therefore be discretized into a finite-dimensional approximation [35].

2.2.3.1 Orthogonal Collocation on Finite Elements

Orthogonal collocation on finite elements (OCFE) is a direct, simultaneous, full discretization approach [34]. The dynamic model of the system, discretized by orthogonal collocation, can be incorporated into the NLP in Equation 2.4.

With OCFE, the prediction horizon is divided into a number of finite elements, and within each element the state trajectories are approximated. In this work, 3rd order Gauss-Radau collocation points are used together with Lagrange interpolation polynomials. For each finite element, the state trajectory is approximated as

$$x_K(t) = \sum_{k=0}^K x_k \ell_k(t) \quad (2.6)$$

where x_k is the state value at collocation point k , and $\ell_k(t)$ is the Lagrange interpolation polynomial which is defined as

$$\ell_k(t) = \prod_{\substack{j=0 \\ j \neq k}}^K \frac{\tau - \tau_j}{\tau_k - \tau_j}, \quad \ell_k(\tau_j) = \begin{cases} 1, & \text{if } j = k \\ 0, & \text{if } j \neq k \end{cases} \quad (2.7)$$

This ensures that the polynomial approximation passes through each collocation point.

The derivatives of the interpolation polynomials evaluated at the collocation points form the differentiation matrix, denoted A_{dot} . Using 3rd order Gauss-Radau collocation points ($K = 3$), which are given for the interval length from 0 to 1, gives

$$A_{\text{dot}} = \begin{bmatrix} -4.1394 & 1.7394 & -3.0000 \\ 3.2247 & -3.5678 & 5.5320 \\ 1.1678 & 0.7753 & -7.5320 \\ -0.2532 & 1.0532 & 5.0000 \end{bmatrix} \quad (2.8)$$

The dynamic constraints are enforced by requiring the polynomial approximation to satisfy the differential equations at each collocation point. This gives

$$A_{\text{dot}}^T \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = dt \cdot f(x_k) \quad (2.9)$$

where $f(x)$ represents the system dynamics (differential equations) evaluated at the collocation points, and dt scale to the finite element time interval.

The value at the first collocation point of the first finite element is known from the initial condition. After solving the collocation equations within one finite element, continuity is enforced by setting the final state of one element equal to the initial state of the next element. In this way, the continuous dynamic system is transcribed into a finite-dimensional NLP problem suitable for MPC.

In this section, the methods used to adapt the optimal inventory control MPC scheme by Kumaraswamy et al. [12] to an ASP will be presented. In addition to inventory (levels) control, the reaction dynamics is taken into account here. Firstly, a description of the dynamic model implemented for tank volumes, biomass and substrate concentrations as well as dissolved oxygen concentration is included. Next, the MPC control formulation is presented detailing all the adaptations made to the original optimal inventory control MPC scheme by Kumaraswamy et al. [12].

3.1 Dynamic Model

In this section, we present the model. We begin with the volume dynamics, thereafter, we introduce the biochemical reaction dynamics, followed by component balances. Next, a simplified settling tank model is described, and the recycle stream is discussed thereafter.

3.1.1 Volume Dynamics

In the MPC scheme, the discrete-time model for tank volumes, V , are

$$V_{t+1} = A \cdot V_t + dt \cdot M \cdot F_t \quad (3.1)$$

Where A is the identity matrix of size $NI \times NI$, and NI is the number of tank inventories. M is the incident matrix of size $NI \times O$ (O is the number of flows), telling which flows enters and leaves each tank. The incident matrix is given in Equation 3.2. dt is the user defined timestep.

$$M_{ij} = \begin{cases} 1 & \text{if } F_j \text{ enters vessel } i, \\ -1 & \text{if } F_j \text{ exits vessel } i, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

3.1.2 Biochemical Reaction Dynamics

The stoichiometric matrix in Table 2.1.1, only describes the internal conversion terms. To describe the component dynamics in a reactor, these conversion rates must be combined with advective terms.

For a completely mixed reactor with volume V , inlet flow rate F_{in} , outlet flow rate F_{out} , inlet concentration $c_{i,in}$, reactor concentration c_i , and conversion rate r_i , the component balance from Equation 2.3 becomes

$$\frac{d(Vc_i)}{dt} = F_{in}c_{i,in} - F_{out}c_i + r_iV. \quad (3.3)$$

Applying this to the three components in Table 2.1.1 gives

$$\begin{aligned} \frac{dX_B}{dt} &= \frac{F}{V} (X_{B,in} - X_B) + \frac{\hat{\mu}S_S}{K_S + S_S} X_B - bX_B, \\ \frac{dS_S}{dt} &= \frac{F}{V} (S_{S,in} - S_S) - \frac{1}{Y} \frac{\hat{\mu}S_S}{K_S + S_S} X_B, \\ \frac{dS_O}{dt} &= \frac{F}{V} (S_{O,in} - S_O) + K_L a (S_O^{sat} - S_O) - \frac{1-Y}{Y} \frac{\hat{\mu}S_S}{K_S + S_S} X_B - bX_B. \end{aligned} \quad (3.4)$$

S_O^{sat} is the saturation concentration of oxygen in the reactor medium. The volumetric mass transfer coefficient, $K_L a$, describes the transfer rate of oxygen from the gas phase into the bioreactor medium. The coefficient is related to the blower power[36]. F denotes the bioreactor inflow. Since we are only considering heterotrophic biomass, we precede with subscript B (instead of BH) for simplicity.

We have used orthogonal collocation on finite elements for the reaction dynamics. The discretization method enables high accuracy approximation, and is effective for stiff systems. In the early stages of this project, we tested the explicit RK4 integration method. This simulation can be seen in Appendix B. However, the RK4 method required a smaller time step, leading to significantly longer computational time. Hence, this was not pursued any further.

The biochemical reaction dynamics, using orthogonal collocation, is shown in Equation 3.5 - 3.8. D is the dilution rate. F_{in} represents the flow entering the bioreactor, and V is the bioreactor volume. Subscript t , represent the current time step, and dt is the step interval. Subscript 1-3 refer to each collocation point, k , and subscript 0, refers to

the initial conditions for each time step. The very first point is known, and the following first collocation points are ensured by continuity. N_t is the prediction horizon.

$$D_t = \frac{F_{in,t}}{V_t} \quad (3.5)$$

$$A_{\text{dot}}^T \begin{bmatrix} X_{B,0,t} \\ X_{B,1,t} \\ X_{B,2,t} \\ X_{B,3,t} \end{bmatrix} = dt \cdot \begin{bmatrix} D_t(X_{B,in,1,t} - X_{B,1,t}) + \mu_{1,t}X_{B,1,t} - bX_{B,1,t} \\ D_t(X_{B,in,2,t} - X_{B,2,t}) + \mu_{2,t}X_{B,2,t} - bX_{B,2,t} \\ D_t(X_{B,in,3,t} - X_{B,3,t}) + \mu_{3,t}X_{B,3,t} - bX_{B,3,t} \end{bmatrix} \quad (3.6)$$

$$\forall t = 0, \dots, N_t$$

$$A_{\text{dot}}^T \begin{bmatrix} S_{S,0,t} \\ S_{S,1,t} \\ S_{S,2,t} \\ S_{S,3,t} \end{bmatrix} = dt \cdot \begin{bmatrix} D_t(S_{S,in,1,t} - S_{S,1,t}) - \frac{1}{Y}\mu_{1,t}X_{B,1,t} \\ D_t(S_{S,in,2,t} - S_{S,2,t}) - \frac{1}{Y}\mu_{2,t}X_{B,2,t} \\ D_t(S_{S,in,3,t} - S_{S,3,t}) - \frac{1}{Y}\mu_{3,t}X_{B,3,t} \end{bmatrix} \quad (3.7)$$

$$\forall t = 0, \dots, N_t$$

$$A_{\text{dot}}^T \begin{bmatrix} S_{O,0,t} \\ S_{O,1,t} \\ S_{O,2,t} \\ S_{O,3,t} \end{bmatrix} = dt \cdot \begin{bmatrix} K_L a_t(S_O^{\text{sat}} - S_{O,1,t}) + D_t(S_{O,in} - S_{O,1,t}) - \frac{1-Y}{Y}\mu_{1,t}X_{B,1,t} - bX_{B,1,t} \\ K_L a_t(S_O^{\text{sat}} - S_{O,2,t}) + D_t(S_{O,in} - S_{O,2,t}) - \frac{1-Y}{Y}\mu_{2,t}X_{B,2,t} - bX_{B,2,t} \\ K_L a_t(S_O^{\text{sat}} - S_{O,3,t}) + D_t(S_{O,in} - S_{O,3,t}) - \frac{1-Y}{Y}\mu_{3,t}X_{B,3,t} - bX_{B,3,t} \end{bmatrix} \quad (3.8)$$

$$\forall t = 0, \dots, N_t$$

3.1.3 Component Balances

The component balances for the non-reactive tanks, using orthogonal collocation, are

$$A_{\text{dot}}^T \begin{bmatrix} X_{B,0} \\ X_{B,1} \\ X_{B,2} \\ X_{B,3} \end{bmatrix} = dt \cdot \begin{bmatrix} \sum M \cdot F_t \cdot X_{B,F,1,t} \\ \sum M \cdot F_t \cdot X_{B,F,2,t} \\ \sum M \cdot F_t \cdot X_{B,F,3,t} \end{bmatrix} \quad \forall t = 0, \dots, N_t \quad (3.9)$$

$$A_{\text{dot}}^T \begin{bmatrix} S_{S,0} \\ S_{S,1} \\ S_{S,2} \\ S_{S,3} \end{bmatrix} = dt \cdot \begin{bmatrix} \sum M \cdot F_t \cdot S_{S,F,1,t} \\ \sum M \cdot F_t \cdot S_{S,F,2,t} \\ \sum M \cdot F_t \cdot S_{S,F,3,t} \end{bmatrix} \quad \forall t = 0, \dots, N_t \quad (3.10)$$

We assume that the outlet concentrations equals the concentration inside the tank, thus the concentration in the flow (subscript F) is collected from tank concentration. The concentrations in the tanks evolve according to concentrations in the inlet of the system and the bioreactor reaction dynamics. We are only considering oxygen concentration inside the bioreactor. Hence, we only account for substrate and biomass concentrations in the non-reactive tanks and flows.

3.1.4 Settling Tank Dynamics

The settling tank is represented using a simplified three-compartment model, as illustrated in Figure 3.1.1. The first compartment does not represent a physical volume, but rather the settling process itself. It receives the effluent from the bioreactor and models the gravitational separation of biomass and clarified water. The incoming flow is divided into two outlet streams: a top stream representing clarified water and a bottom stream representing settled sludge.

The top stream enters the top compartment, which represents the clarified-water zone of the settling tank. It is assumed that this stream contains 10% of the biomass and substrate concentrations entering the settler, corresponding to $0.1X_B$ and $0.1S_S$. The bottom stream enters the bottom compartment, which represents the sludge blanket and receives the remaining 90% of the biomass and substrate, corresponding to $0.9X_B$ and $0.9S_S$.

The flow split is assumed constant and is described by

$$F_{\text{Top}} = \frac{1}{3}F_{\text{Bioreactor}}, \quad F_{\text{Bottom}} = \frac{2}{3}F_{\text{Bioreactor}}. \quad (3.11)$$

This approach provides a simplified representation of solids separation while avoiding the complexity of detailed settling dynamics. Since the first compartment is solely a

symbolic representation of the separation process, its volume is constrained to remain constant and cannot be depleted during the simulation.

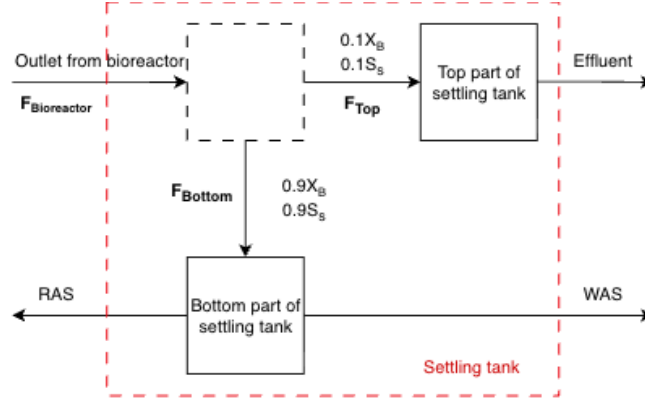


Figure 3.1.1: Settling tank model.

3.2 Control Formulation

3.2.1 Development of the MPC Formulation

The paper by Turan et al. [11] proposed an MPC formulation to optimally control bottleneck shifting for inventories in a sequential topology. The formulation was further developed by Kumaraswamy et al. [12], to be applicable to general process topologies. This has been achieved by introducing weighting factors to outflows. The general MPC problem, extended with reaction dynamics and component balance, is shown in Equation 3.12.

$$\begin{aligned}
 \min_x \quad & J \\
 \text{s.t.} \quad & V(t_{k+1}) = AV(t_k) + MF(t_k) \\
 & f(x) = 0 \\
 & 0 \leq F(t_k) \leq F_{max} \\
 & 0 \leq K_L a \leq K_L a_{max}
 \end{aligned} \tag{3.12}$$

$f(x)$ refer to the biochemical reaction dynamics and the component balances, in Equation 3.5-3.8 and Equation 3.9 - 3.10. F_{max} is a vector including the maximum limits for each flow. A is the identity matrix of volumes. M is the incident matrix. $K_L a$ is the volumetric mass transfer coefficient for oxygen transfer from gas to liquid. F and $K_L a$ are input variables.

The original objective function by Turan et al. [11], is implemented such that outflow and volumes are maximized:

$$J = - \sum_{k=0}^{N_k} \gamma^k \left(F_N(t_k) + \sum_{i=1}^{N_I} \alpha_i^{\mathcal{I}} V_i(t_k) \right) \quad (3.13)$$

$\alpha_i^{\mathcal{I}}$ is the weighting factor of tank volume. γ^k is a time discount factor, and it is included to avoid multiple solutions for the MPC[11].

Kumaraswamy et al. [12] further developed the objective, to adapt it for general process topologies:

$$\min J = - \sum_{k=0}^{N_k} \gamma^k \left(\underbrace{\sum_{j \in \mathcal{O}} \alpha_j^{\mathcal{O}} F_j(t_k)}_{\text{System Outflow Maximization}} + \underbrace{\sum_{i=1}^{N_I} \alpha_i^{\mathcal{I}} V(t_k)}_{\text{Inventory Maximization}} + \underbrace{P}_{\text{Penalty term}} \right) \quad (3.14)$$

The extended cost function can be applied to process systems with more complex network structures, including flow splits, flow mergers, and recycle streams [12]. The formulation incorporates weighting factors on system outflows and penalty terms for constraint handling. An algorithm was developed to determine the inventory weights based on user-specified edge weights and outflow weighting factors, thereby establishing the relative priority of each tank within the process network. Thus, the MPC can prioritize inventory units and outflows consistent with the objective of maximizing throughput.

In this study, we want to achieve optimal inventory control while satisfying operational and process constraints. Thus, an MPC is designed seeking throughput maximization and optimal inventory control, but only without exceeding pollutant limits and without draining the bioreactor volume. This is because pollutant removal is the most important goal for an ASP. The inventory control MPC scheme presented by Kumaraswamy et al. [12] can be incorporated to such systems by adapting their weight assignment strategy. For all tanks except the bioreactor, we assign weights to maximize tank volumes such that we can utilize their buffer capacities. In other words, these tanks are drained before throughput maximization is compromised. As a result, we can isolate the bottleneck and prevent disturbance propagation to essential units. In the case of an ASP, we assume that maintaining the volume of the aerated bioreactor is more important than throughput maximization, meaning that the outflow should be reduced instead of reducing the bioreactor volume.

In many cases there are multiple feasible solutions to the MPC. In order to avoid frequent changes in the control strategy and achieve smooth control, penalization terms are included in the cost function. Soft constraints has been assigned penalty terms of varying weight. By this, it becomes clear to the MPC how to achieve optimal control of the system. Operational constraints are typically more important than the optimization

objective and should therefore be assigned a larger penalty[12].

3.2.2 The Volumetric Mass Transfer Coefficient, K_La

As shown in Equation 2.2, dissolved oxygen is utilized during biomass growth and decay. Consequently, the dissolved oxygen concentration may decrease as the biological activity increases. Although the dissolved oxygen concentration does not affect biomass growth in the simplified model, we want to include control of the dissolved oxygen concentration. This is because it is a common control aspect in aerated bioreactors. To maintain a sufficient dissolved oxygen concentration for microbial growth and decay, the volumetric mass transfer coefficient, K_La , is included as a manipulated variable in the optimization problem. In practice, K_La is not manipulated directly. Rather, the aeration intensity is adjusted, resulting in changes to the coefficient. In this work, K_La is used as the manipulated variable as a simplified representation of aeration control.

The dissolved oxygen concentration is controlled through the implementation of a quadratic state tracking term in the objective function,

$$Q_{S_O} \left(S_O - S_O^{\text{setpoint}} \right)^2 \quad (3.15)$$

where S_O^{setpoint} is the desired dissolved oxygen concentration and Q_{S_O} is the penalty for deviations from this setpoint.

3.2.3 Return Activated Sludge

The recycle stream in the ASP is used to return flocculated biomass from the settling tank to the aerated bioreactor, thereby maintaining a high biomass concentration in the reactor. A high biomass concentration increases the availability of active microorganisms, which enhances the removal of organic matter. Thereby, the overall treatment efficiency improves. There are several factors affecting treatment efficiency in an ASP, such as influent flow rate, influent wastewater characteristics, temperature, etc. The recycle flow rate also affect the system performance, and is regarded as one of the more convenient factors to control[37]. There are three typical control strategies: Constant RAS Flow Rate, Constant RAS Ratio, and Sludge Blanket Level Control[37]. In the constant RAS ratio strategy, the influent flow rate is measured and the RAS flow rate is adjusted to maintain a constant RAS ratio. The ratio is defined as the RAS flow rate divided by the influent flow rate[37]. In practice, the flow rate of the recycle stream depends upon system characteristics. Metcalf & Eddy, Inc. [38] state that the ratio should be between 100-150% of influent flow. We therefore use a setpoint of 1.25 for the recycle ratio. This is included as a tracking term in the cost function:

$$(R - R^{\text{setpoint}})^2 \cdot Q_R \quad (3.16)$$

The ratio is

$$F_{RAS} = R \cdot F_{in} \quad (3.17)$$

If we choose to regulate the recycle ratio, it would be conflicting to simultaneously maximize the outflow WAS . Therefore, we do not assign an outflow maximization weighting factor to the WAS flow. Instead, the MPC choose its flow rate according to the objectives and constraints in the optimization problem.

3.2.4 MPC Formulation

For each unique process, it is important to understand how the objective function should be constructed with weighting factors and penalties, in order to achieve desired controller response. In this work, we do not consider that throughput maximization is always equivalent with maximizing economics, and thus the algorithm for weight assignment of inventories is modified. However, we use the objective function in Equation 3.14, and manually assign weights to inventories, according to the process requirements. In addition, we include quadratic setpoint tracking terms for the controlled variables - dissolved oxygen concentration S_O and the recycle ratio R .

$$\begin{aligned} \min J = & - \sum_{k=0}^{N_k} \gamma^k \left(\underbrace{\sum_{j \in \mathcal{O}} \alpha_j^{\mathcal{O}} F_j(t_k)}_{\text{System Outflow Maximization}} + \underbrace{\sum_{i=1}^{N_I} \alpha_i^{\mathcal{I}} V(t_k)}_{\text{Inventory Maximization}} + \underbrace{P}_{\text{Penalty term}} \right) \\ & + \sum_{k=0}^{N_k} Q_{S_O} \left(S_O - S_O^{\text{setpoint}} \right)^2 + \sum_{k=0}^{N_k} Q_R \left(R - R^{\text{setpoint}} \right)^2 \end{aligned} \quad (3.18)$$

The term P includes penalization of soft constraint violations. In the MPC scheme, a limit on outlet substrate concentration and maximum and minimum volumes for each tank are included as soft constraints. The penalization term is structured as following

$$\begin{aligned} P = & \sum P_1 \cdot S_{S\text{slack}} \\ & + \sum P_2 \cdot V_{\text{slack}_{\text{bioreactor}}} + \sum P_2 \cdot V_{\text{surplus}_{\text{bioreactor}}} \\ & + \sum P_3 \cdot V_{\text{slack}_{\text{non-reactive tanks}}} + \sum P_3 \cdot V_{\text{surplus}_{\text{non-reactive tanks}}} \end{aligned} \quad (3.19)$$

P with subscript 1-3, are penalty weights, with decreasing magnitude. $S_{S\text{slack}}$ is the concentration deviation from $S_{S,\text{max}}$. $V_{\text{slack}_{\text{bioreactor}}}$ is the volume deviation from V_{max} of the bioreactor, and $V_{\text{surplus}_{\text{bioreactor}}}$ is the volume deviation from V_{min} . $V_{\text{slack}_{\text{non-reactive tanks}}}$ and $V_{\text{surplus}_{\text{non-reactive tanks}}}$ are the deviation from maximum and minimum volume

limits for the non-reactive tanks. The relative magnitudes of the penalties establish a hierarchy among the soft constraints. Consequently, the MPC prioritizes satisfaction of the substrate concentration constraint, followed by the bioreactor volume constraints, while the volume constraints of the non-reactive tanks are assigned the lowest priority.

3.2.5 Closed-Loop Simulation Disturbances

Flow rate and inlet substrate concentration disturbances are introduced to evaluate the disturbance rejection capabilities of the NMPC. The disturbances are not known a priori by the optimizer and are therefore not included in the prediction model. Instead, the controller responds to the disturbances based on the current measurements and the dynamic model.

The bottleneck shift scenarios are introduced to evaluate the inventory control strategy under changing operating conditions. A reduction in the maximum influent flow rate represents periods of reduced wastewater generation or upstream flow limitations. The disturbance forces the controller to utilize the buffer capacities, and maintain process operation while satisfying constraints and maximizing throughput.

The influent substrate concentration disturbance is introduced to represent variations in organic load in wastewater treatment plants. Such variations may arise from diurnal patterns, seasonal shifts, etc. A step increase in influent substrate concentration is applied, allowing evaluation of the controller's ability to satisfy substrate concentration constraints.

We assume that each state is measurable, there are no measurement noises, and there are no model mismatch. There has not been conducted an economic analysis of the process. Furthermore, we have not considered hydraulic or sludge retention time. In addition, we have not considered temperature and pH effects on the system.

3.2.5.1 Model parameters

The parameter values applied in the simulations are collected in 3.2.1. The parameters Y , $\hat{\mu}$ and b has subscript H . This is because it is parameters for heterotrophic biomass. In the reference Henze et al. [16], they include all 19 parameters for the ASM1, but in this text, we only consider heterotrophic biomass growth and decay.

Table 3.2.1: Model parameters and typical domestic wastewater characteristics (Denmark, 20°C)

Parameter	Value	Unit	Reference
Y_H	0.67	g cell COD formed (g COD oxidized) ⁻¹	Henze et al. [16]
$\hat{\mu}_H$	6.0	day ⁻¹	Henze et al. [16]
K_S	20.0	g COD m ⁻³	Henze et al. [16]
b_H	0.62	day ⁻¹	Henze et al. [16]
S_O^{sat}	10.0	g O ₂ m ⁻³	Nelson et al. [39]

The inlet concentrations of substrate, biomass and oxygen are given in Table 3.2.2.

Table 3.2.2: Inlet values

Parameter	Value	Unit	Reference
$S_{S,in}$	200	g COD m ⁻³	Nelson et al. [39]
$X_{BH,in}$	0	g COD m ⁻³	Nelson et al. [39]
$S_{O,in}$	2.0	g O ₂ m ⁻³	Nelson et al. [39]

CASE STUDIES

This section presents the case studies conducted in this work. The case studies are arranged in order of increasing complexity to ensure satisfactory controller performance on simpler systems before considering more complex process configurations. The simplest configuration, consisting of a single bioreactor, is presented first. Subsequently, the bioreactor is incorporated into a series of tanks. Finally, a modified activated sludge process is considered. In case 1-3, the component balance for concentration in non-reactive tanks has not been included. The component balance has been introduced in case 4, when the recycle stream is included.

For all case studies, the simulations are initialized with reactor volumes at their maximum values and all flow rates at steady state. The same model parameters, listed in Table 3.2.1, are used throughout the cases. Influent concentrations of X_B , S_S , and S_O are taken from Table 3.2.2. Each simulation is run for 10 days with step time of 1 hour and a prediction horizon of 1 day.

The following sections begin with an introduction to the case and its objectives, followed by a description of the applied perturbations and an interpretation of the resulting system behavior.

4.1 Case 1: Single Bioreactor Tank

In this case, the system consists of a bioreactor with in- and outflow. The system is illustrated in Figure 4.1.1. We include this example, to show how the MPC manages the trade-off between inventory and outflow maximization. In addition, this example illustrates how the dynamics of state variables X_B , S_S , and S_O are coupled. The controller objective is to maximize outflow and volume, while keeping the substrate concentration within a maximum value ($S_{S, max}$) of 10 gCOD/m³.

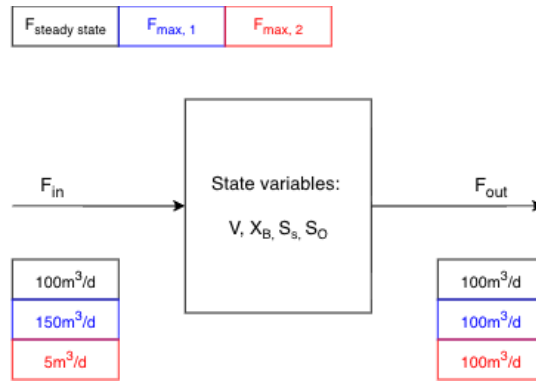


Figure 4.1.1: Block diagram of single bioreactor system

In the simulation, we consider that the bottleneck initially is located at the outflow, F_{out} . The maximum flow rates ($F_{max, 1}$ in Figure 4.1.1) are $150\text{m}^3/\text{d}$ and $100\text{m}^3/\text{d}$ for inflow and outflow, respectively. Shortly after 2.5 days, the bottleneck shifts to the inlet. The maximum flow rate of F_1 is reduced to $5\text{m}^3/\text{d}$ ($F_{max, 2}$). The initial steady state flow ($F_{steady\ state}$) for both inlet and outlet is $100\text{m}^3/\text{d}$. For this case, K_La is kept constant with a value of 120 d^{-1} . Maximum volume is 1000m^3 and the minimum volume is 970m^3 , which are included as a soft constraints in the MPC. Outflow maximizing is weighted more heavily than volume maximizing in the MPC objective function.

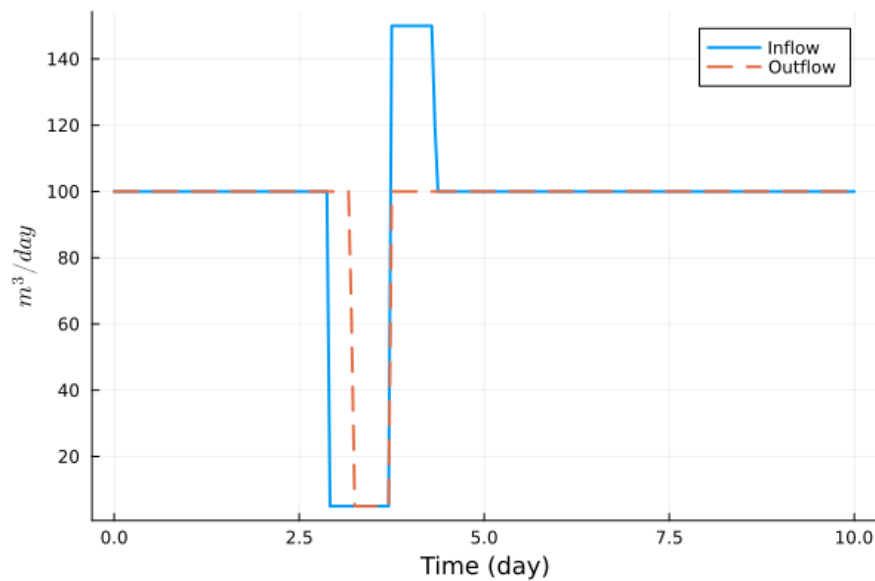


Figure 4.1.2: Flow rates in Single Bioreactor system.

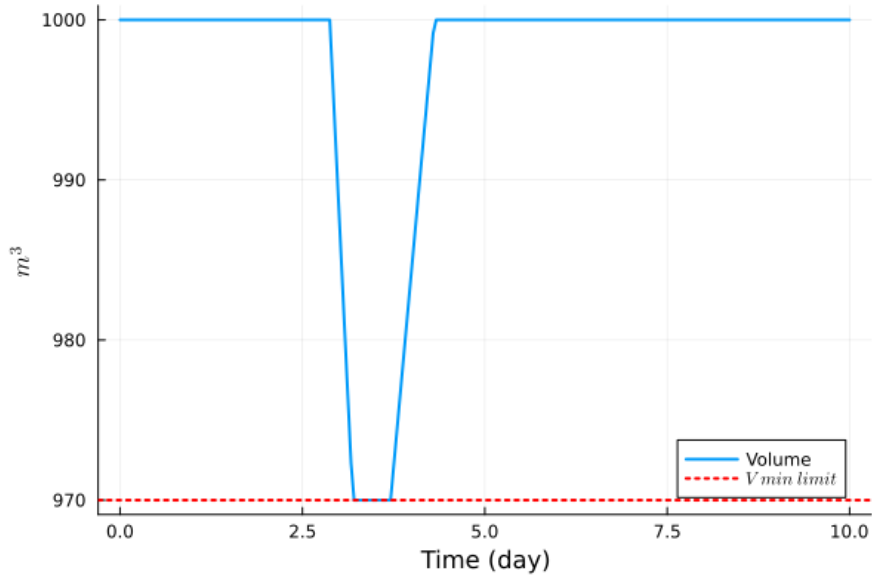


Figure 4.1.3: Volume in bioreactor

Figure 4.1.2 shows the bottleneck shift that occurs shortly after 2.5 days when the maximum inflow rate is reduced. The controller maintains the outflow at $100\text{m}^3/\text{d}$, causing a reduction in volume, shown in Figure 4.1.3. As the volume approaches its minimum limit, the outflow is reduced to prevent constraint violation. This behavior is consistent with the MPC objective function, where maximizing outflow is prioritized over reactor volume, while violations of the volume constraints are heavily penalized.

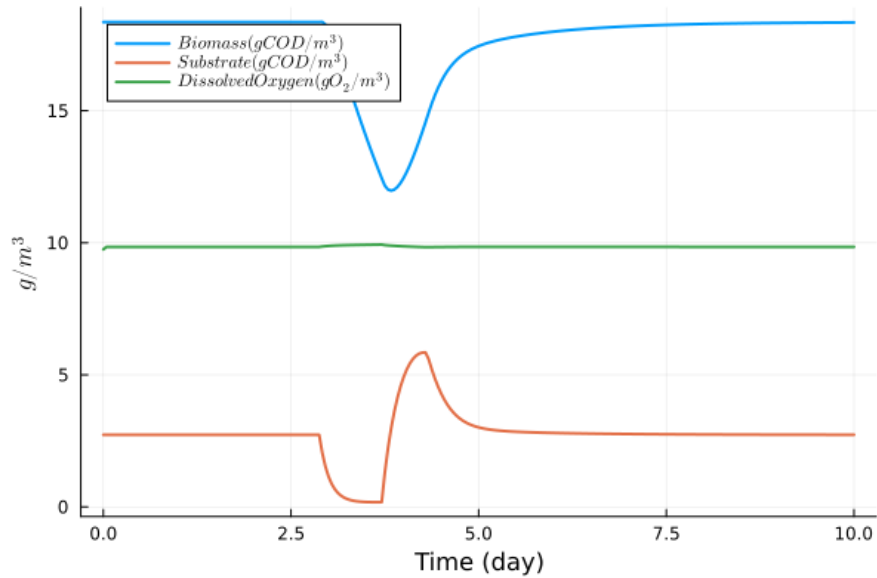


Figure 4.1.4: State variables in bioreactor

Figure 4.1.4 shows the concentration states of the bioreactor (X_B , S_S , S_O). The dynamics are governed by the dilution rate, the inlet concentrations, and the reaction rates, see Equation 3.5 - 3.8. Initially, biomass, substrate and dissolved oxygen concentrations

are at their steady state values. When the inflow is reduced, the advective contribution in the substrate balance decreases. At the beginning of the simulation, the biomass concentration remains high, so substrate consumption through the Monod term remains significant. As a result, substrate is consumed faster than it is supplied, leading to the rapid decrease in S_S . The decrease in substrate also reduces the biomass growth rate. Consequently, the biomass concentration X_B decreases, since growth is no longer sufficient to compensate for decay and dilution.

When the flow rate increases again, substrate is supplied to the reactor, causing S_S to rise. We observe that the substrate concentration is kept far under maximum value (10 gCOD/m³), thus the concentration constraint is not active in this case. The rise in substrate concentration promotes biomass growth, which explains the subsequent increase in X_B . When volume is returned to its maximum value, and the flow rates approaches their steady state values, both S_S and X_B gradually return toward their steady state values.

The dissolved oxygen concentration S_O remains comparatively stable because the oxygen transfer term dominates the oxygen balance. The oxygen transfer coefficient, K_La , is maintained at a sufficiently high and constant value throughout the simulation, while the driving force ($S_O^{sat} - S_O$) continuously promotes oxygen transfer into the liquid phase. Although oxygen consumption varies with the biological growth rate, the oxygen transfer term compensates for these changes, resulting in minor deviations in S_O .

4.2 Case 2: Bioreactor Incorporated into a Sequential Tank System

In this case, the system consists of three serially connected tanks, with the bioreactor serving as the middle tank. The system is presented in Figure 4.2.1. We include this example to verify that the MPC correctly manages inventories in a system containing multiple tanks. We assume that there is no reaction taking place in tank 1 and 3. Furthermore, concentrations in flows and non-reactive tanks has not been considered.

The controller objective is to maximize outflow and volumes, while keeping the substrate concentration within a maximum value ($S_{S, max}$) of 10 gCOD/m³. Weights are assigned in the cost function such that outflow is given the highest priority, followed by the bioreactor volume, the volume in Tank 3, and finally the volume in Tank 1. Maximum and minimum constraints on the tank volumes, as well as the maximum substrate concentration constraint, are included in the MPC as soft constraints. Substrate concentration violation is penalized more than volume violations. The penalties are included in the cost function.

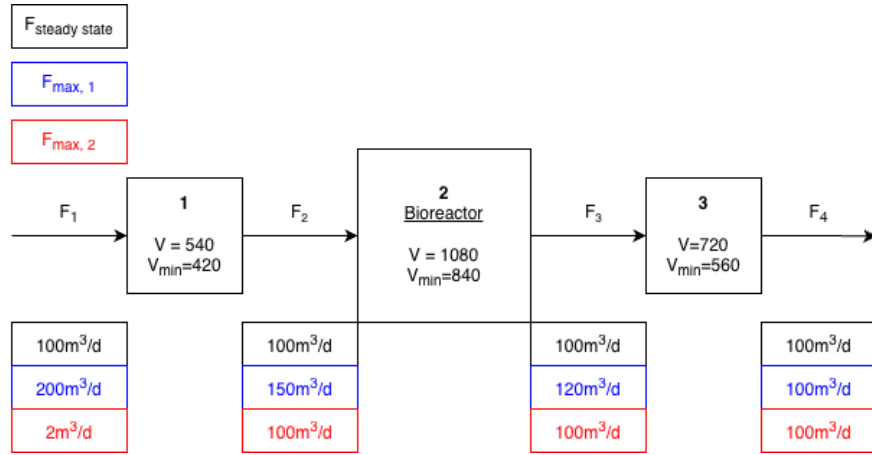


Figure 4.2.1: Block diagram of bioreactor in a sequential arrangement.

In the simulation, we consider that the bottleneck initially is located at the outflow, F_4 . The maximum flow rates ($F_{max, 1}$) are 200, 150, 120, and $100\text{m}^3/\text{d}$ for flows 1 through 4, respectively. After 5 hours, the bottleneck shifts to the inflow. The maximum flow rate of F_1 is reduced to $2\text{m}^3/\text{d}$ ($F_{max, 2}$). The initial steady state flow rate ($F_{steady\ state}$) for each flow is $100\text{m}^3/\text{d}$. For this case, K_La is kept constant with a value of 120d^{-1} . The maximum volumes are 90% of tank capacity, whereas the minimum volumes are 70% of tank capacity which corresponds to 420, 840, 560 m^3 for tanks 1 through 3, respectively. (The minimum volumes are set high to reduce simulation time.)

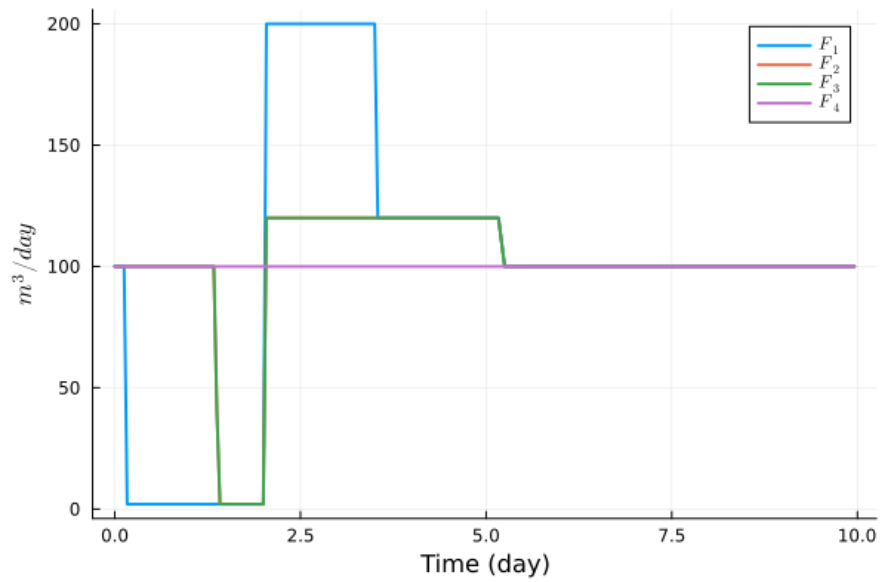


Figure 4.2.2: Flow rates.

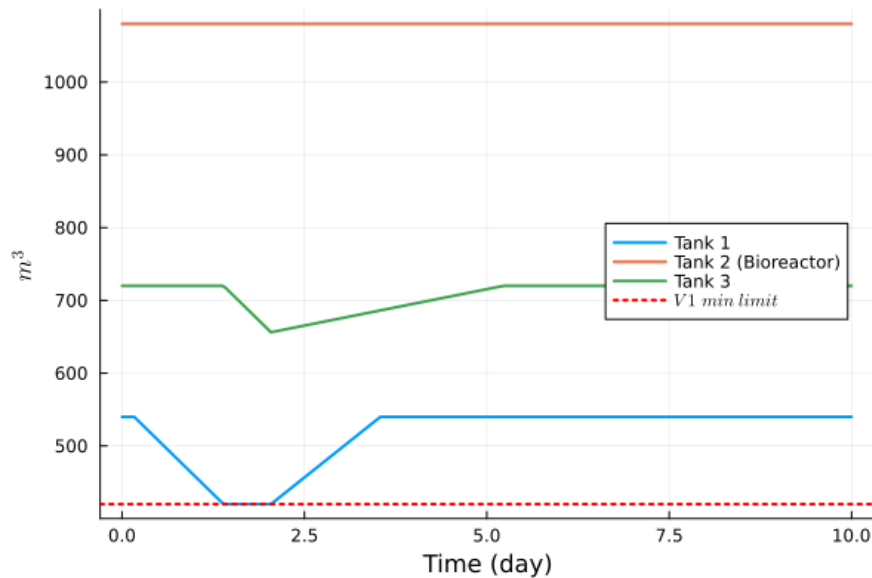


Figure 4.2.3: Volumes.

Figure 4.2.2 illustrates the bottleneck shift that occurs after 5 hours when the maximum inflow rate F_1 is reduced. The controller maintains the remaining flows at $100\text{m}^3/\text{d}$, causing volume 1 to be decreased, shown in Figure 4.1.3. When tank 1 reaches its lower volume limit, both flows F_2 and F_3 is being reduced so that the bioreactor volume is kept at maximum value. Since the outflow F_4 is kept constant through out the simulation, the volume of tank 3 is reduced. This result shows that the MPC regulates the inventories corresponding to the assigned weights in the objective function.

Shortly before 2.5 days, the bottleneck shifts back to the outflow. To satisfy the objective function, F_1 is maximized to increase the volume in Tank 1. Since the bioreactor volume is already at its upper limit, F_2 cannot exceed F_3 without violating the maximum volume constraint. Consequently, both F_2 and F_3 are maintained at $120\text{ m}^3/\text{d}$, corresponding to the maximum flow rate of F_3 , even though F_2 is capable of reaching $150\text{ m}^3/\text{d}$. The increased flow rate subsequently enables the volume in Tank 3 to increase. The overall response is consistent with the MPC objective function.

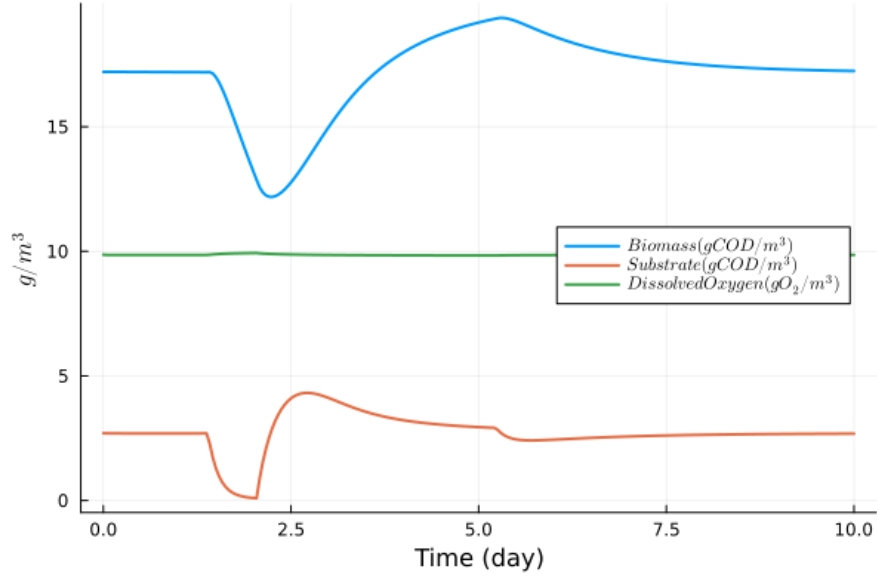


Figure 4.2.4: State variables, X_B , S_S , S_O .

For the concentration states in Figure 4.2.4, we observe similar results as in Case 1. However, we see that substrate and biomass is closer to reaching steady state between 2.5 and 5 days. This is due to longer duration of flow perturbation.

4.3 Case 3: Sequential Tank System with Substrate Perturbation

We consider the same system as before, which is depicted in Figure 4.3.1. In this case, we perturb the system not only with a flow disturbance, but also with an increased inlet substrate concentration. In biological secondary treatment processes, dissolved oxygen concentration may become a limiting factor for biological substrate removal. Thus, S_O is typically a controlled variable, with a set point between 2-4g/m³. In order to maintain S_O at set point, the aeration is regulated. We want to include the regulation of dissolved oxygen concentration, even though the biological processes are not limited by S_O in our model. As the volumetric mass transfer coefficient is related to blower power, K_{La} is selected as the manipulated variable in the MPC. This represents a simplification in which changes in blower power are represented by corresponding changes in K_{La} , and the blower dynamics are therefore not modeled explicitly.

We include this example to verify that the MPC regulates inventories corresponding to given weights and satisfy process constraints, with the presence of additional control tasks and disturbances. The controller objective is to maximize outflow and volumes, while keeping dissolved oxygen concentration at setpoint (3g/m³), and the substrate concentration within a maximum value of 15 gCOD/m³. The seriatim amongst inventories in a decreasing priority order is outflow F_4 , volume of bioreactor, followed by volume of tank 3.

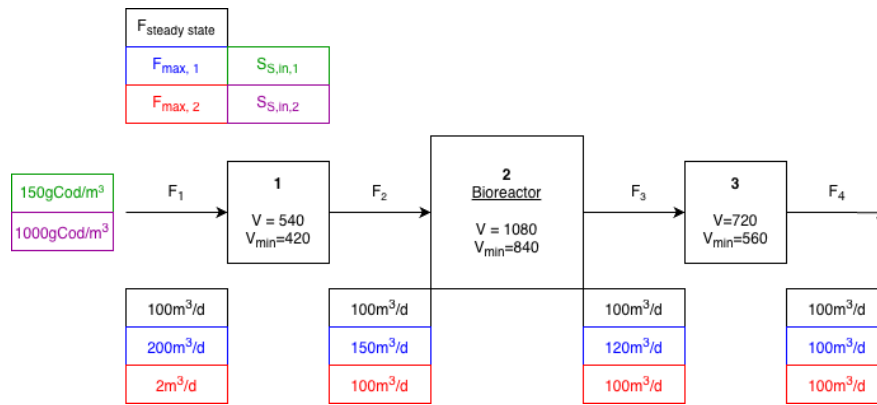


Figure 4.3.1: Block diagram of bioreactor in a sequential arrangement, with substrate perturbation.

In the simulation, we consider that the bottleneck initially is located at the outflow, F_4 . The maximum flow rates ($F_{\max, 1}$) are 200, 150, 120, and $100 \text{ m}^3/\text{d}$ for flows 1 through 4, respectively. After 20 hours, the bottleneck shifts to the inflow. The maximum flow rate of F_1 is reduced to $2 \text{ m}^3/\text{d}$ ($F_{\max, 2}$). The initial steady state flow rate ($F_{\text{steady state}}$) for each flow is $100 \text{ m}^3/\text{d}$. Shortly after 6 days, the system is disturbed by a significant increase of substrate inlet concentration ($150 \rightarrow 1000 \text{ gCOD}/\text{m}^3$). The minimum volumes are 70% of tank capacity which corresponds to 420, 840, 560 m^3 for tanks 1 through 3, respectively.

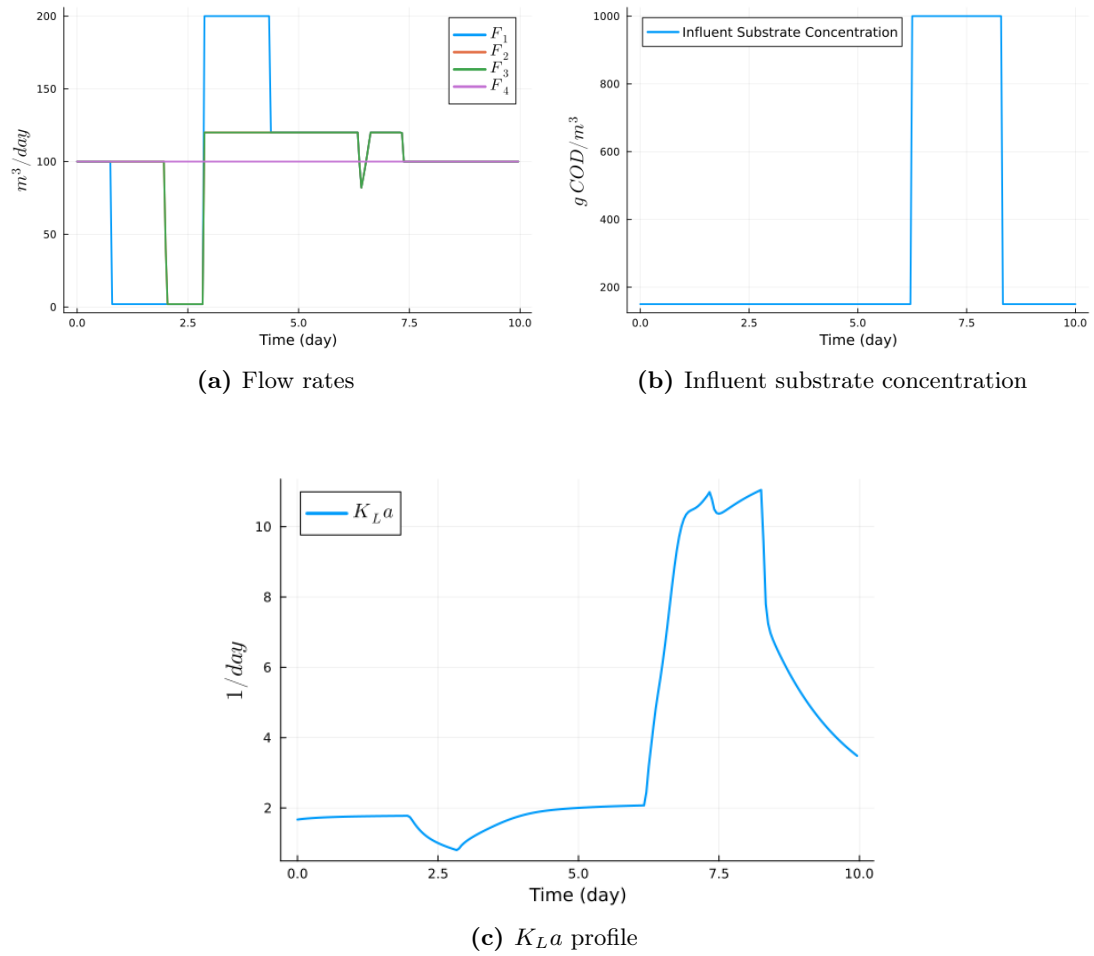


Figure 4.3.2: Flow rates, influent substrate concentration profile, and $K_L a$ profile.

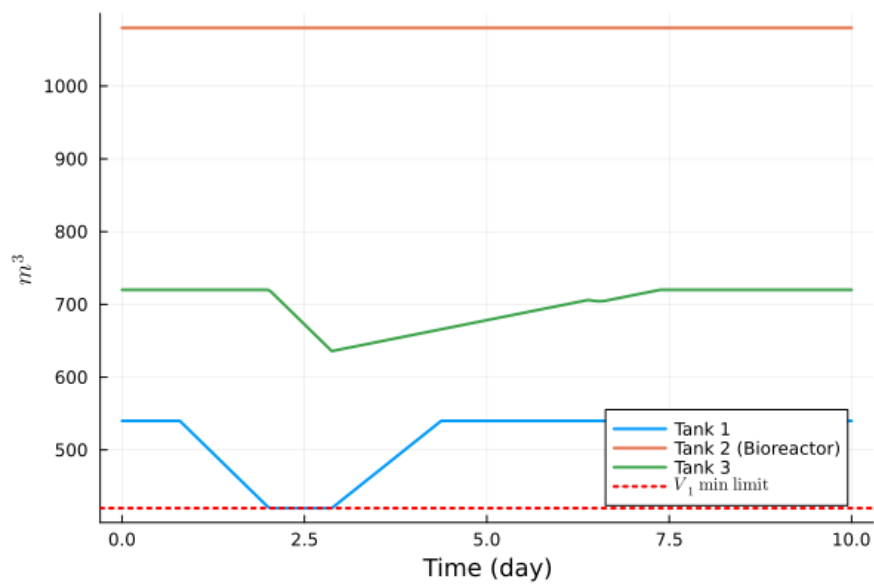


Figure 4.3.3: Volumes.

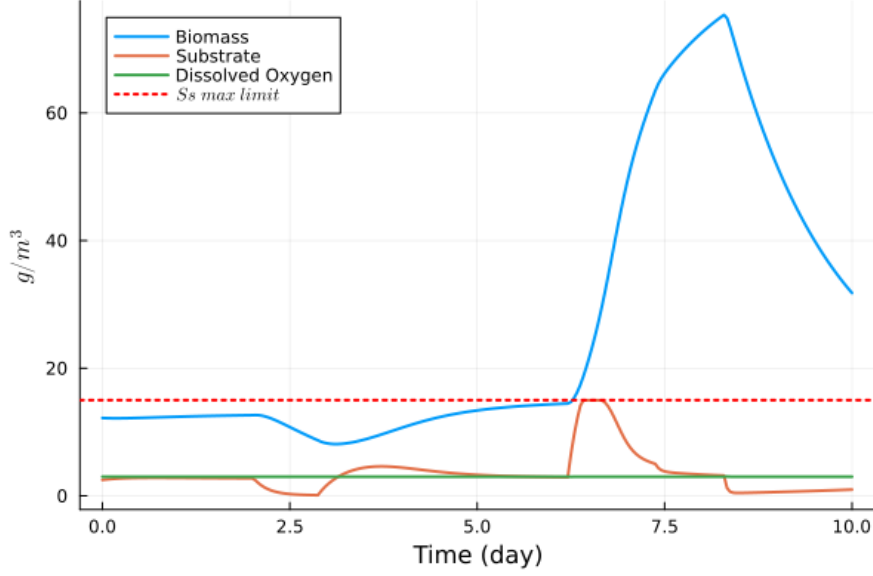


Figure 4.3.4: State variables, X_B , S_S , S_O .

We observe similar results for the state variables as in Case 2, until the inlet substrate concentration increase. However, in this case, the dissolved oxygen concentration is maintained at its set point of 3 g/m^3 by manipulating K_{La} . As shown in 4.3.2c, K_{La} decreases as the biomass and substrate concentrations decrease and increases as they rise. In other words, as the oxygen demand decreases, a lower K_{La} is required to maintain the dissolved oxygen concentration at its set point, and vice versa.

Shortly after 6 days, the system is disturbed by a significant increase in the influent substrate concentration. As a result, the substrate concentration in the bioreactor rapidly approaches its maximum allowable value. To prevent violation of the substrate concentration constraint, the MPC temporarily reduces the bioreactor inflow and outflow. By reducing the throughput, the incoming substrate load is limited while the bioreactor volume remains constant. Consequently, the increase in Tank 3 volume is temporarily halted. As the substrate concentration is brought below its constraint limit, the flow rates (F_2 and F_3) return to their previous values and the filling of Tank 3 resumes. Furthermore, K_{La} increases significantly during this period. The elevated substrate concentration promotes biomass growth, which increases the oxygen demand in the bioreactor. The MPC therefore increases K_{La} to maintain the dissolved oxygen concentration at its setpoint.

Overall, the MPC regulates inventories and satisfies constraints according to given weights and penalties.

4.4 Case 4: Activated Sludge Process

In this section we will present a modified activated sludge process which is illustrated in Figure 4.4.1. The system consists of one mixing tank (tank 1), the bioreactor (tank 2)

and a settling tank (tank 3, 4, 5). Tank 3 does not correspond to a physical tank and it is solely a representation of the separation process inside the settling tank. Hence, the volume in Tank 3 cannot be reduced. The recycle stream of activated sludge is included. We assume that no reactions occur in either the mixing tank or the settling tank. However, component balances are included for flows and tanks as the recycle flow is implemented.

We include this example to investigate the controllers performance on the ASP. The controller objective is to maximize outflow and volumes, while keeping dissolved oxygen concentration and the recycle ratio at setpoint ($3\text{g}/\text{m}^3$ and 1.25 , respectively), and the substrate concentration within a maximum value of $15\text{ gCOD}/\text{m}^3$. In this case, the inventories and outflows are weighted in the following order: volume of bioreactor, outflow F_4 , recycle flow F_7 , tank volume 4, tank volume 5, tank volume 1, with the bioreactor volume having the highest priority. K_La and flow rates are a manipulated variables.

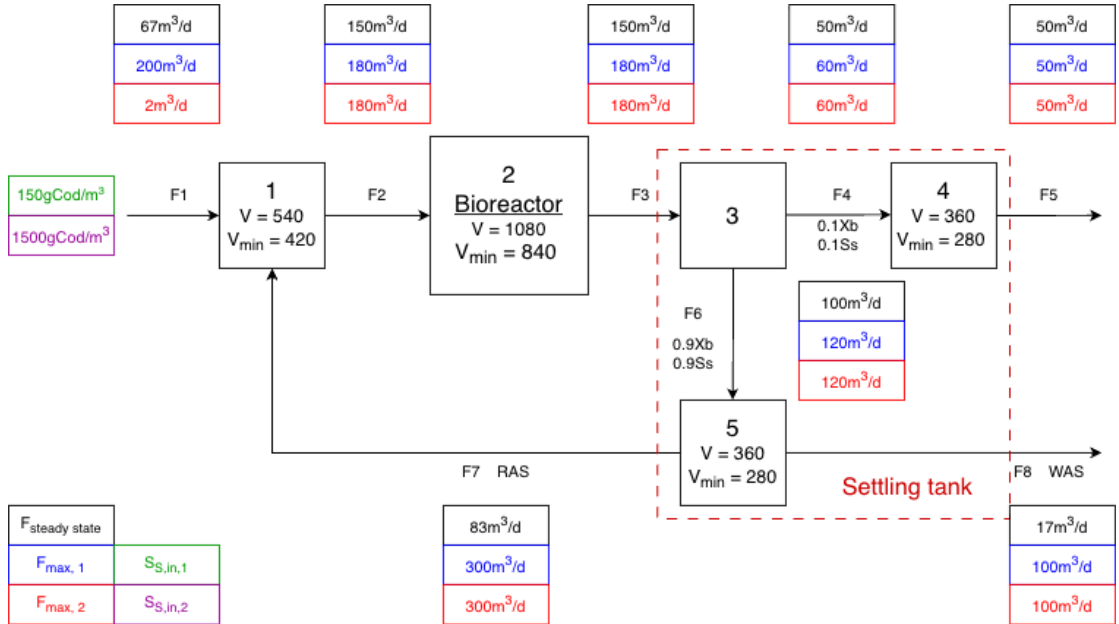


Figure 4.4.1: Block diagram of activated sludge process.

For this case, we have performed two simulations with different disturbances. This is to enable a more systematic verification of the controller performance. Firstly, we present the case with a flow disturbance, followed by a step change in inlet concentration.

4.4.1 Flow disturbance

In the simulation, we consider that the bottleneck initially is located at the outflow, F_4 . The maximum flow rates ($F_{max, 1}$) are 200, 180, 180, 60, 50, 120, 300, 100m³/d for flows 1 through 8, respectively. After 20 hours, the bottleneck shifts to the inflow. The maximum flow rate of F_1 is reduced to 2m³/d ($F_{max, 2}$). The initial steady state flow rate ($F_{steady state}$) for each flow is 67, 150, 150, 50, 50, 100, 84, 17m³/d for flows 1 through

8, respectively. The minimum volumes are 70% of tank capacity which corresponds to 420, 840, -, 280, 280m³ for tanks 1 through 5, respectively.

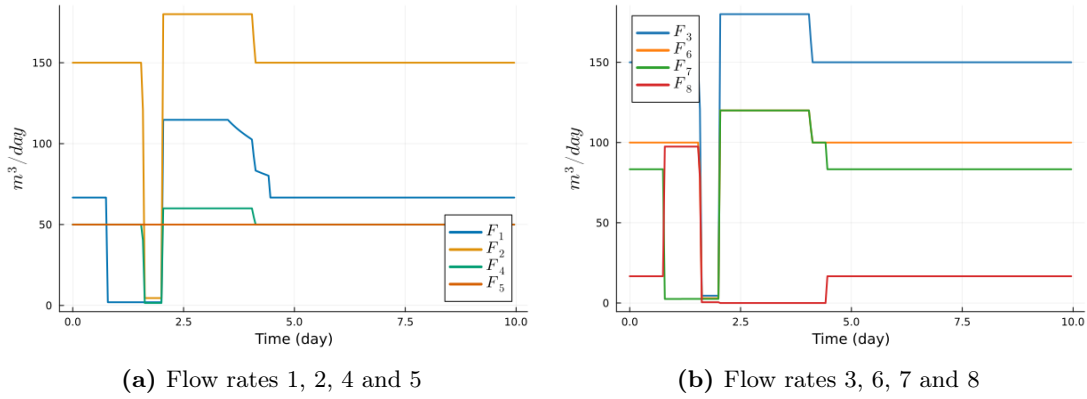


Figure 4.4.2: Flow rates.

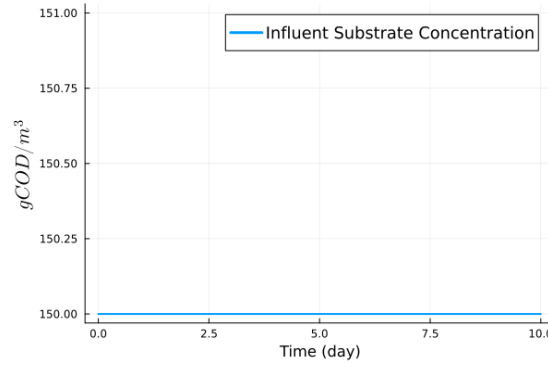


Figure 4.4.3: Constant inlet substrate concentration.

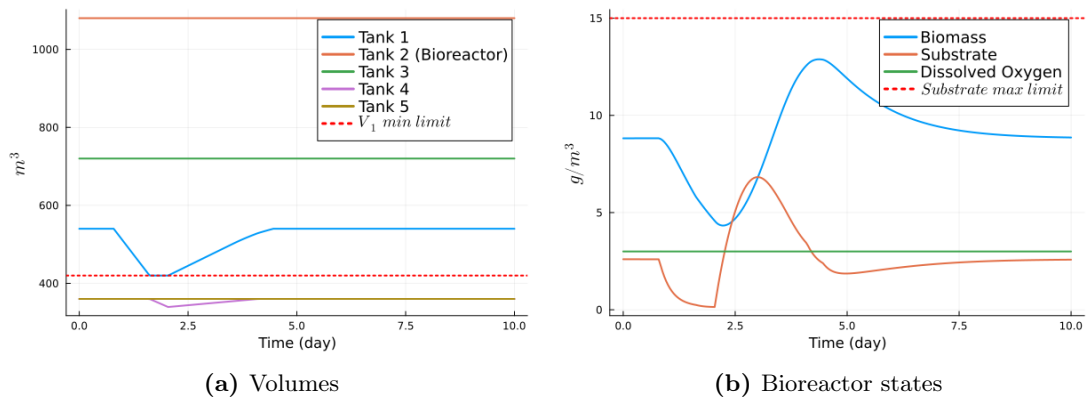


Figure 4.4.4: Tank volumes and bioreactor states.

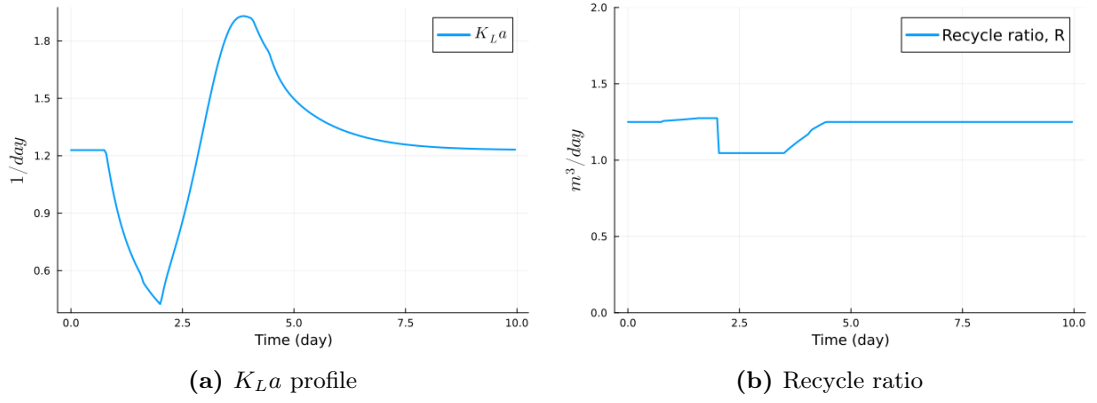
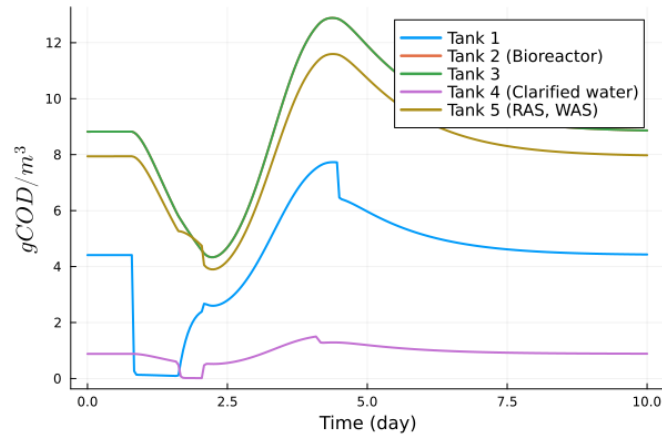
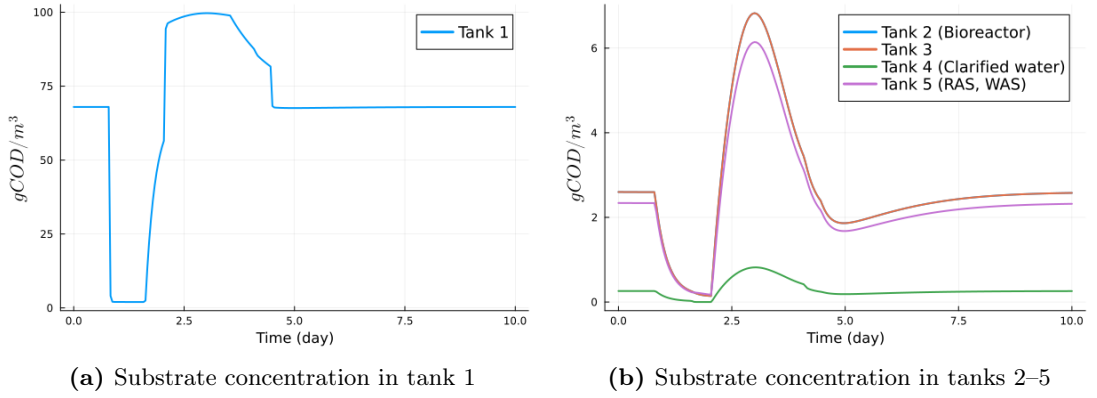
Figure 4.4.5: $K_L a$ and recycle ratio.

Figure 4.4.6: Substrate and biomass concentrations in the tanks.

1. Steady state

- These results are explained sequentially, we start looking at the plots from steady state at 0 days, towards 10 days. Here, S_S , X_S and S_O refers to the concentrations in the bioreactor.

2. **Shift of bottleneck:** $F_{1,max} = 200\text{m}^3/\text{d} \rightarrow F_{2,max} = 2\text{m}^3/\text{d}$.

- F_1 is reduced.
- Consequently, F_7 is reduced to avoid significant deviation from the recycle ratio setpoint.
- The recycle ratio, $R = F_7/F_1$, is slightly increased as F_1 is reduced. Since the MPC simultaneously seeks to maximize tank volumes, F_7 is not reduced proportionally to F_1 .
- F_8 is increased as F_7 is reduced, this is to avoid flooding in tank 5.
- Each remaining flow is maintained constant.
 - $\Rightarrow$ The volume in tank 1 decreases, as F_1 and F_7 are decreased while F_2 is maintained constant.
 - $\Rightarrow$ The remaining tank volumes are maintained constant.
- The minimum volume limit in tank 1 is reached shortly after 1.5 days.
- F_2 is reduced to avoid volume violation in tank 1.
- F_3 is reduced according to the reduction of F_2 , as the volume of the bioreactor is the highest priority in the MPC cost function.
- F_4 and F_6 are reduced proportionally to F_3 . This is because the volume of tank 3 cannot be reduced, as it represents the separation process of the settling tank.
- F_5 is maintained constant, as it has the second-highest priority in the cost function.
- F_8 is reduced as F_6 is reduced, in order to avoid depletion of tank 5.
 - $\Rightarrow$ Volume in tank 4 decreases, as F_5 is maintained constant while F_4 is reduced.
 - $\Rightarrow$ The recycle ratio R stabilizes as a new steady state is reached with the volume in tank 1 at its minimum limit.
- S_S decreases, see Figure 4.4.4b. This is because of the dilution effect caused by the increased recycle flow rate and reduced inflow to tank 1. Since the biomass concentration initially remains at its previous steady state value, substrate consumption remains high. The combination of reduced substrate input and sustained biomass consumption leads to a rapid decline in substrate concentration.
- X_B decreases as a consequence of substrate reduction. The biomass decay becomes more dominant than biomass growth.

- K_La is reduced with decreasing X_B as the dissolved oxygen demand declines. See Figure 4.4.5a.
- K_La is manipulated to successfully keep S_O at setpoint, $3 \pm 0.0007\text{g/m}^3$.

3. **Shift of bottleneck:** $F_{2,max} = 2\text{m}^3/\text{d} \rightarrow F_{1,max} = 200\text{m}^3/\text{d}$.

- F_1 is increased above its initial steady state flow rate to recover the volume in tank 1.
- F_2 and F_3 are increased above their initial steady state flow rates to recover the volume in tank 4.
- Consequently, both F_4 and F_6 are increased according to their ratio to F_3 .
- Outflow F_5 is maintained constant at maximum flow rate.
- F_7 is increased to maximum possible flow rate ($120\text{m}^3/\text{d}$, which is maximum flow rate of F_6), to support volume recovery in tank 1.
- The recycle ratio is reduced, as the maximum flow rate of F_7 is too low to maintain the setpoint value of 1.25. The MPC prioritizes to maximize volume over keeping R constant, which corresponds to weighting factors and penalties in the objective function.
- Consequently, F_8 is reduced to zero.
 $\Rightarrow$ Volume in tank 4 is maximized. Flow rates F_2 , F_3 , F_4 , F_6 and F_8 are returned to their initial steady state flow rates. F_1 and F_7 are reduced similarly, but not returned to their initial steady state values yet.
 $\Rightarrow$ Volume in tank 1 is maximized. F_1 and F_7 are returned to their initial steady state flow rates.
- S_G increases as a result of increased throughput of the bioreactor. The substrate concentration passes the biomass concentration, as biomass growth is proportional to substrate concentration in a mixed order.
- X_B increases until S_G is balanced by consumption. Thereafter, the biomass concentration decreases slightly as the system moves toward steady state.
- K_La is increased and subsequently stabilized according to biomass growth, such that S_O remains at setpoint.

The biomass and substrate concentrations in the non-reactive tanks evolve according to the flow dynamics and the concentration dynamics in the bioreactor.

This case demonstrates that the MPC achieves satisfactory performance. Although tank 4 is weighted more heavily than tank 5, volume in tank 4 is reduced first. This behavior arises from a trade-off where the optimizer avoids severe constraint violation of minimum volume in tank 1, significant deviation from the setpoint of recycle ratio, and

volume maximizing. In addition, this solution does not affect the outflow, as outflow F_4 is maintained at maximum flow rate. Furthermore, after bottleneck shift, the MPC achieves tank filling.

4.4.2 Influent substrate concentration perturbation

Now, we perturb the system with an increased inlet substrate concentration. The initial steady state flow rate ($F_{steady\ state}$) for each flow is 67, 150, 150, 50, 50, 100, 84, 17 m³/d for flows 1 through 8, respectively. The minimum volumes are 70% of tank capacity which corresponds to 420, 840, -, 280, 280 m³ for tanks 1 through 5, respectively. Shortly after 3 days, the system is disturbed by an increase of substrate inlet concentration, from 150 to 1500 gCOD/m³.

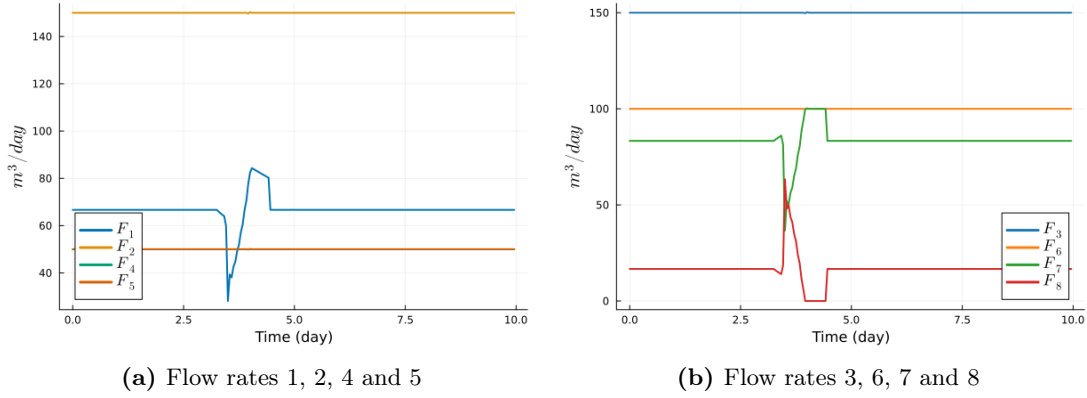


Figure 4.4.7: Flow rates.

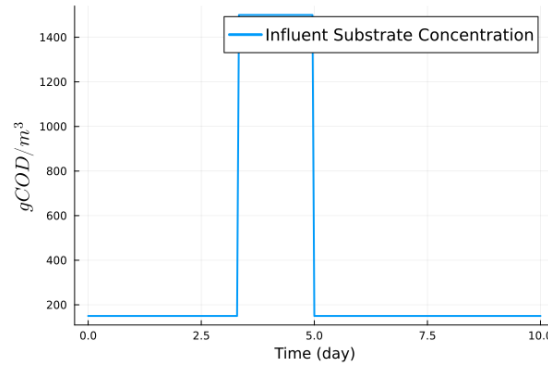


Figure 4.4.8: Inlet substrate concentration.

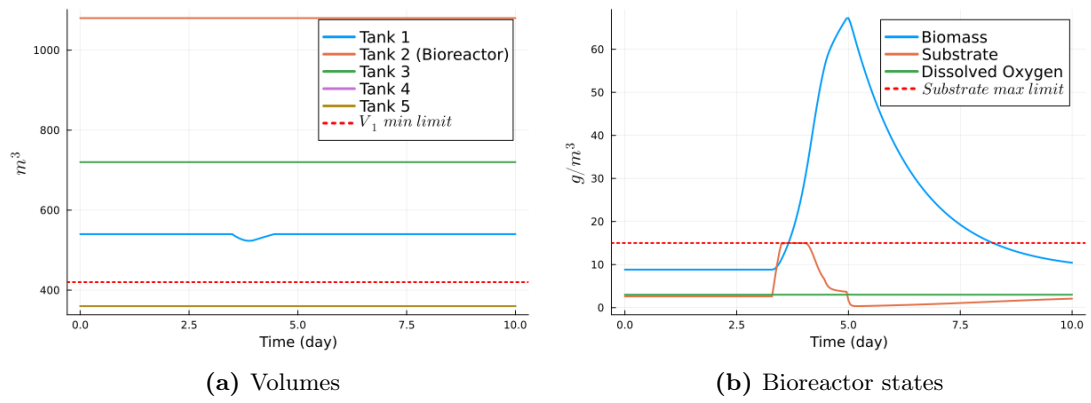


Figure 4.4.9: Tank volumes and bioreactor states.

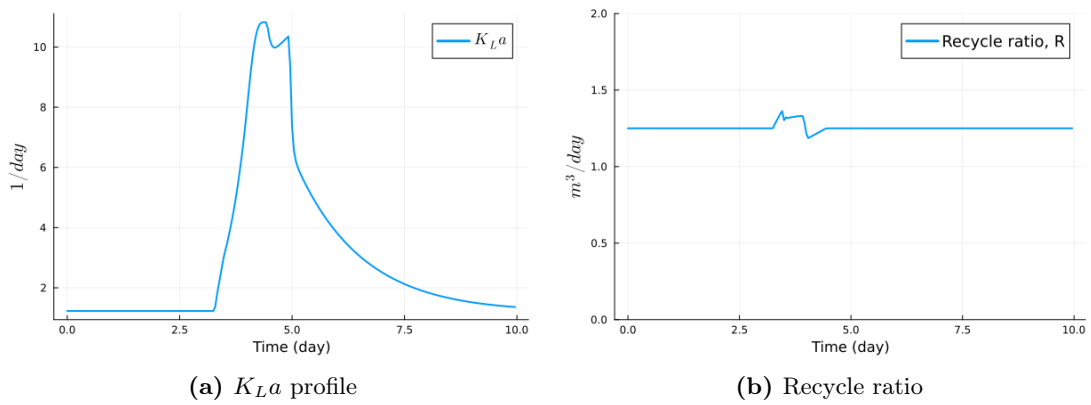


Figure 4.4.10: K_La and recycle ratio.

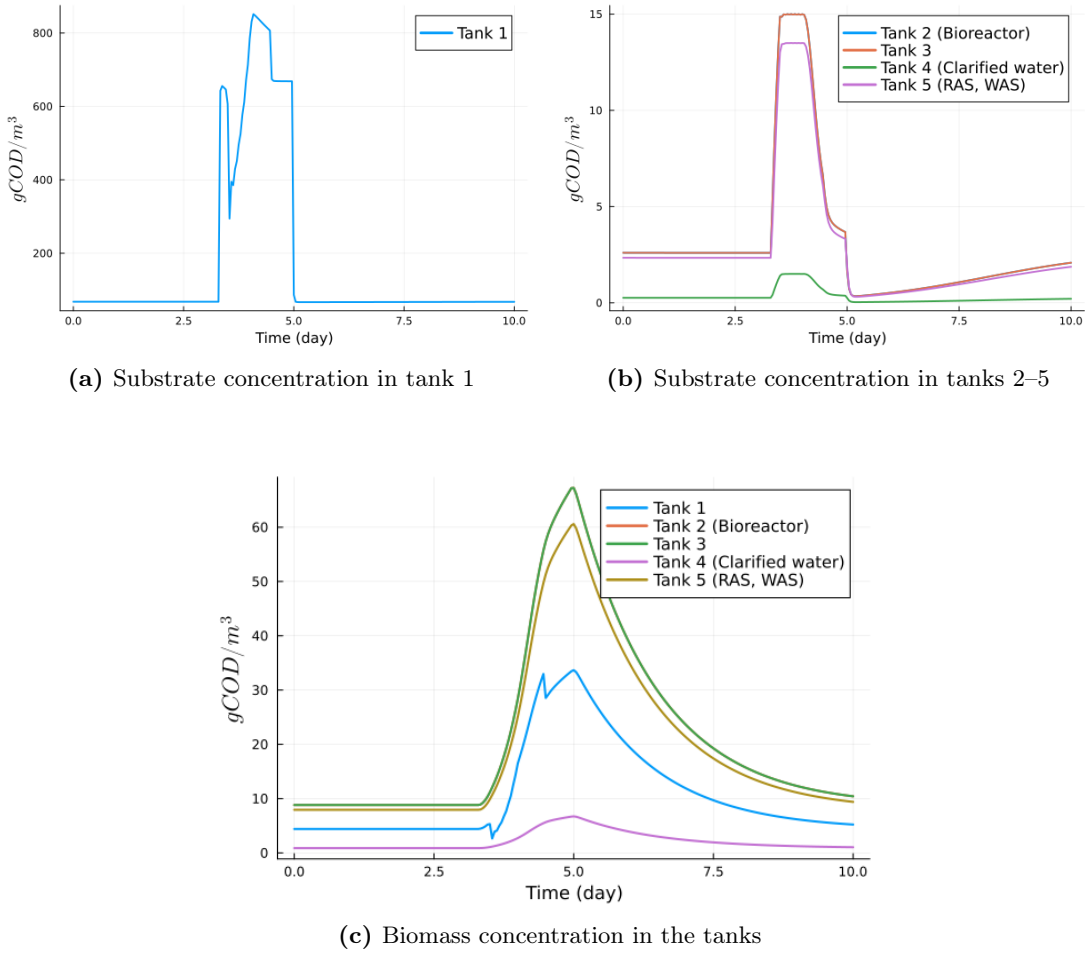


Figure 4.4.11: Substrate and biomass concentrations in the tanks.

1. Steady state

- These results are explained sequentially, we start looking at the plots from steady state at 0 days, towards 10 days. Here, S_S , X_S and S_O refers to the concentrations in the bioreactor.

2. Disturbance: Inlet substrate concentration increases from 150 $gCOD/m^3 \rightarrow 1500$ $gCOD/m^3$ at 5 days.

- S_S rapidly increases as a consequence of the increased influent substrate concentration and approaches its upper limit.
 $\Rightarrow F_1$ is reduced in order to reduce the substrate load entering the system.
 $\Rightarrow F_7$ is reduced according to the recycle ratio setpoint, $R = F_7/F_1 = 1.25$. However, R is slightly increased. This has a diluting effect on the substrate concentration in tank 1. Thus, this also contributes to reduce the inlet substrate concentration into the bioreactor. By this, the MPC prevents violation of the substrate concentration constraint.
 $\Rightarrow F_8$ is increased in order to prevent flooding in tank 5.
 $\Rightarrow$ The volume in tank 1 decreases as F_1 and F_7 are reduced.

- Flow rates F_2 , F_3 , F_4 , F_5 and F_6 is maintained constant. By this, the remaining volumes are kept at their maximum limit, and the outflow F_5 is maximized.
- X_B increases as the increased substrate availability promotes biomass growth.
- As biomass grows, substrate consumption increases. The increased biomass concentration prevents further accumulation of substrate and maintains S_S below its upper limit.
 $\Rightarrow F_1$ is increased to recover the volume in tank 1. Hence, F_7 is increased to keep R close to setpoint. However, R is slightly below setpoint, as F_7 is limited by the flow rate of F_6 . Consequently, F_8 is reduced to zero.
 $\Rightarrow$ After volume recovery, F_1 , F_7 and F_8 return to their steady state values.
- K_La is increased as a result of the increased oxygen demand associated with biomass growth.
 $\Rightarrow$ The dissolved oxygen concentration is maintained at its setpoint throughout the simulation.

3. **Disturbance:** Inlet substrate concentration decreases from 1500 gCOD/m³ $\rightarrow$ 150 gCOD/m³.

- All flow rates are at steady state.
- S_S rapidly decreases because substrate concentration into the bioreactor is reduced while the biomass concentration remains high, sustaining a high substrate consumption rate.
- X_B decreases as decay becomes the governing process when substrate availability is reduced.
 $\Rightarrow S_S$ start to increase. The system searches for steady state.
- S_O is regulated by K_La .
- K_La is decreased as a result of lower dissolved oxygen demand with biomass decay.

The biomass and substrate concentrations in the non-reactive tanks evolve according to the flow dynamics and the concentration dynamics in the bioreactor.

This case demonstrates that the optimizer exploits the liquid inventory of tank 1 to avoid violation of maximum substrate limit.

Through these case studies, the MPC demonstrates satisfactory performance. The controller successfully manages inventories, while satisfying operational constraints. Furthermore, the MPC is computationally efficient.

DISCUSSION AND FUTURE WORK

This study demonstrates that MPC can effectively achieve optimal inventory control while simultaneously managing process constraints for a simplified ASP model. The optimal inventory control MPC formulation proposed in subsection 3.2.1 was successfully extended to a more realistic process representation. To achieve this, a multivariable dynamic model consisting of reaction and component balances was integrated into the MPC framework. This extension is significant because many chemical and process systems are inherently multivariable, and an inventory control strategy must account for relevant operational constraints to be practically applicable. For the ASP modeled in this study, the MPC exhibited behavior consistent with the control objectives while respecting operational constraints. However, the study is based on several assumptions and contains limitations that provide avenues for future work.

One limitation of this work is the use of a highly simplified activated sludge model. The model only considers biomass, substrate, and dissolved oxygen, while neglecting several important biological processes included in ASM1, such as nitrification, denitrification, and multiple biomass fractions. Although this simplification reduces computational complexity and facilitates a preliminary verification of controller performance, it limits the model's ability to accurately represent the behavior of a real secondary treatment process. Future work should therefore extend the modeling framework to more detailed activated sludge models in order to evaluate whether the proposed inventory control strategy remains effective under more realistic process conditions.

Another limitation is the simplified representation of the settling tank. The settling process is modeled using fixed flow splits and fixed separation of components between the clarified water and sludge compartments. In practice, settling performance depends on factors such as influent flow rate and settling time, which may vary over time. Consequently, the current model does not accurately capture the settling tank dynamics.

Future work should incorporate a more detailed settling tank model to better represent sludge separation dynamics and assess its influence on controller performance.

Furthermore, the aeration system has not been modeled. Instead, the volumetric mass transfer coefficient, $K_L a$, is used directly as a manipulated variable to control the dissolved oxygen concentration. In practice, the manipulated variable is the power supplied to the aeration equipment. Moreover, the type of aeration has not been considered, although different technologies, such as diffused aeration and mechanical aeration, exhibit different oxygen transfer efficiencies. Consequently, future work could incorporate a more detailed model of the aeration system to provide a more realistic control formulation.

Other assumptions associated with ASP modeling are as follows. In the presented cases, we assume that the concentration inside the bioreactor is equivalent with the outlet concentration. Furthermore, we assume that there are no reactions occurring upstream or downstream of the bioreactor. These are not realistic assumptions, and should be taken into account in the development of the ASP dynamic model.

In the activated sludge process, we control the return activated sludge flow by constant ratio to the system inflow. In contrast, the waste activated flow has no constraints, and can therefore be manipulated towards zero flow rate. In real applications, the waste activated sludge is implemented to remove excess biomass, avoid significant changes in sludge characteristics and the sludge retention time. In other words, the waste flow is important, as it is an available manipulated variable which could be used to control the sludge retention time. This could be incorporated within the MPC cost function.

A key assumption in this work is that all state variables are measurable and available to the controller. In practice, several biological variables in wastewater treatment plants, such as biomass concentration, are not necessarily measured continuously. As a result, the controller performance demonstrated in this study may be optimistic compared to real industrial implementation. Future work could therefore investigate the use of state estimation techniques, to estimate unmeasured states from available process measurements.

The disturbance scenarios considered in this work are applied in the shape of step change. In practice, influent flow rates and wastewater characteristics are subject to uncertainty and may vary significantly due to weather conditions and diurnal patterns. Consequently, the controller performance observed in the simulations may not fully represent operation under realistic disturbances. Future work should evaluate the controller under more realistic disturbance shapes. An alternative is a sinusoidal shape, that follow typical diurnal patterns.

Further improvements could be made to the cost function implemented in the MPC. The objective function emphasizes inventory management, throughput, and process constraint satisfaction, but does not explicitly account for economics. In particular, aeration energy and pumping costs for the recycle flow are not considered. As a result, the con-

troller cannot guarantee economically optimal operation. Future work could extend the formulation toward an economic MPC, where operating costs and treatment objectives are incorporated directly into the optimization problem.

Lastly, we want to emphasize the importance of weighting factors and penalties. The selection of appropriate weighting and penalty factors is crucial for achieving satisfactory MPC performance. These factors must be chosen appropriately to clearly communicate the relative importance of various control objectives to the optimizer. As a result, we avoid solution multiplicities and the numerical difficulties arising when the optimization problem exhibits a weakly defined optimum.

CONCLUSIONS

This thesis investigated the application of model predictive control for optimal inventory management in a simplified activated sludge process for wastewater treatment. The primary objective was to extend an existing optimal inventory control framework to a process containing reaction dynamics and associated constraints.

To achieve this, a dynamic model of a simplified ASP was developed, incorporating biomass, substrate, and dissolved oxygen dynamics. The model was integrated into a nonlinear MPC formulation using orthogonal collocation on finite elements. The controller was designed to balance inventory management, throughput maximization, and process constraints, while regulating dissolved oxygen concentration and return activated sludge flow rate.

A series of case studies with increasing complexity was used to evaluate controller performance. The results demonstrated that, by an appropriate selection of weighting factors and penalties, the MPC successfully managed inventories and satisfied operational constraints, under both flow disturbances and influent substrate concentration disturbances. In the activated sludge process, the controller optimally utilized available buffer capacity, maintained substrate concentrations within specified limits, and regulated dissolved oxygen and the recycle flow. Overall, the proposed framework showed satisfactory performance within reasonable computational load.

The work demonstrates that the optimal inventory control MPC framework can be extended to systems with reaction dynamics and recycle streams, representing an important step towards applying the MPC scheme in real process plants. For practical implementation of this MPC, future work should focus on the following key aspects. Firstly, more detailed models for the activated sludge process and settling tank need to be incorporated. Secondly, aeration and sludge management should be explicitly accounted for. Economic MPC formulations that account for aeration and pumping costs can be inves-

tigated to improve the profitability delivered by the controller. Lastly, state estimation must be implemented for unmeasured state variables.

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APPENDICES

A - GITHUB REPOSITORY

Code files used in this document are included in the Github repository linked below.
The work has been solved with Ipopt in Julia environment.

Github repository link

- <https://github.com/anelycke/Master-Thesis#master-thesis>

B - SIMULATION OF BIOREACTOR WITH RK4

We present a simulation of the bioreactor with RK4 as integration method. The prediction horizon is 3 days and step time is 1 hour. Parameters and inlet values used from Table 3.2.1 and 3.2.2, and K_{La} is constant at 120 d^{-1} . The simulation results in steady state behavior. However, the simulation completed in approximately 25 minutes, which is computationally demanding considering that this is a pure simulation with no optimization.

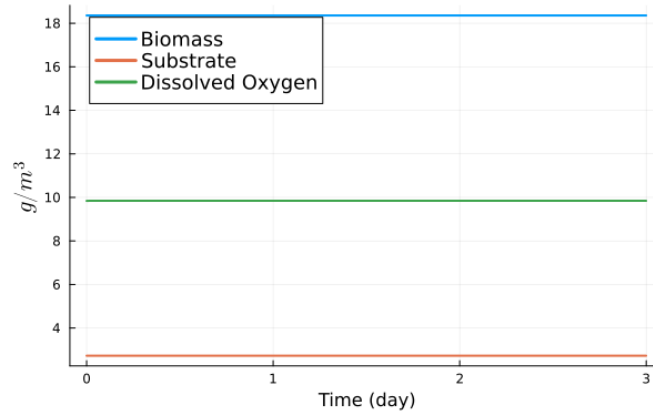


Figure .0.1: Components in bioreactor.

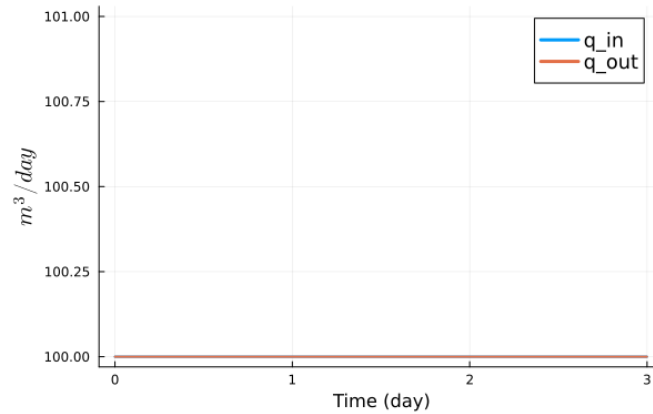


Figure .0.2: Flow rates.